# Supplementary Information

Metric random tilings: exact entropy and the diffraction of counted disorder

*v3.0b patch (2026-11-11): paracrystal disambiguation sync with the main text — S10.2 and Table S1 updated (continuous lattice-vector distribution, average local orientation, $\psi_6 < 1$, entropy not countable); no numbers changed.*

*v3.0c patch (2026-11-11): erratum on the entropy-split ideal-mixing reference (Sect. S11 and Fig. S11) — the honeycomb counting torus carries E = 3F edges at F hexagonal faces (Sect. S1.2), so the ideal alloy-mixing reference is (E/F) ln k = 3 ln k per face (2.08 / 3.30 for k = 2 / 3), not 1.5 ln k as previously computed and plotted; the closure-cost readings double accordingly (2.14 at F = 6, 2.49 at F = 15). Measured entropies, exact counts and the entropy-split identity are unchanged; `si_entropy_split.csv` and Fig. S11 regenerated from the corrected enumerator. Script file-name pointers in S4, S6 and S7 updated to the released (ASCII) engine filenames.*

## S1. Model register: dictionary, closure conditions, naming, and boundaries

### *S1.1 The dictionary and the missing row*

Periodic tilings (space groups) describe crystals; Penrose-type aperiodic tilings describe quasicrystals; random tilings of fixed shapes carry the combinatorial entropy of random-tiling theory[S1]; rule-free random networks describe amorphous solids. In all four rows the order (or its absence) is combinatorial. The row added here holds the combinatorics fixed (one vertex type, one face type) and lets the *metric* (the edge lengths) fluctuate under exact local closure rules.

### *S1.2 Equiangular closure conditions*

With all interior angles fixed, the face-closure constraints are homogeneous linear equations in the edge lengths (Fig. S1a). For an equiangular hexagon with edge lengths $L_0 \ldots L_5$ taken cyclically (directions at 60° steps), closure $\Sigma\, L_j u_j = 0$ is equivalent to two linear equations; in the indexing convention of our scripts, $\mathbf{L_1+L_2 = L_4+L_5}$ **and** $\mathbf{L_0+L_5 = L_2+L_3}$ (opposite-side sum rules)[S2]. An equiangular triangle is equilateral (all three edges equal); an equiangular quadrilateral is a rectangle (opposite sides equal); equiangular octagons in the 4.8.8 network have directions mixing 1 and √2/2 (Sect. S5). The alphabet plays three roles at once: values, Wang matching colours, and geometric lengths (Fig. S1b). A single tile already carries a four-dimensional family of closed shapes at fixed angles: the "one hexagon, one module" intuition of the rigidity-and-deficiency criterion (Fig. S1c).

*Torus counting construction.* The counting torus is the quotient of the honeycomb by the sublattice $L_x\, t_1 + L_y\, t_2$ generated by two primitive translations: two vertices (sublattices A, B) per fundamental cell $(i, j) \in \mathbb{Z}^2/(L_x, L_y)\mathbb{Z}^2$, three edges per cell (A(i,j)–B(i,j) within the cell, A(i+1,j)–B(i,j) in the +x

direction, A(i,j+1)–B(i,j) in the +y direction, all indices closed modulo ($L_x$, $L_y$)), and one hexagonal face per cell, F = $L_x \cdot L_y$. Each face contributes its six edge slots in counter-clockwise boundary order; every edge of the quotient is one slot of each of its two adjacent faces and carries a **single shared value** (the gluing). The closure constraints are the two slot equations above on every face. This identification, the sheared (affine) fundamental domain drawn in Fig. 3a, is the domain of every count in the paper; other primitive-pair choices (e.g. the standard shear-free embedding) give a different quotient at finite size, binary 3×3: Ω = 518 (affine) versus 842 (standard), and the convention is fixed throughout. In code, edges are keyed by their endpoint vertex triples, which keeps the two parallel edges of a two-cell domain distinct (S6.4).

*Worked smallest tori.* The construction reproduces the archived anchors exactly; with E = 3F and $dim_{\mathbb{R}}$ = F + 2 verified by Fraction-exact rank elimination at every size (including the resonant ones, S6.3):

| Torus | class | F | E | $dim_{\mathbb{R}}$ | resonant? | closed form | Ω (exact) |
|---|---|---|---|---|---|---|---|
| 2×2 | binary {1,2} | 4 | 12 | 6 | yes (2 ≡ 0 mod 2) | $2^5 = 32$ | 64 = 2 × 32 |
| 3×2 | binary {1,2} | 6 | 18 | 8 | no | $2^6 = 64$ | 64 (bitwise) |
| 3×3 | binary {1,2} | 9 | 27 | 11 | yes | $2^7 = 128$ | 518 |
| 2×2 | ternary {1,2,3} | 4 | 12 | 6 | yes | — | 729 |
| 3×2 | ternary {1,2,3} | 6 | 18 | 8 | no | — | 1017 |
| 3×3 | ternary {1,2,3} | 9 | 27 | 11 | yes | — | 22 767 |

Each Ω is the digit-for-digit agreement of two independent code paths (DFS enumeration over the slot equations, and the transfer-matrix trace; S11), with brute force checking every alphabet assignment against the two slot equations on the smallest rows. The 2×2 binary row already shows the resonant excess (64 = 2 × closed form), the smallest instance of the winding sectors quantified in S6.3.

### *S1.3 Scope of the Wang-type invocation*

The tiling counterpart of the metric random tiling ensemble is a Wang-type tiling[S3] used in its permissive regime only: a finite tile set, local edge-matching rules that are exactly realizable, extensibility (configurations exist at arbitrary sizes), and a positive entropy density. We make no claim involving aperiodicity forcing or undecidability; the signature results of the classical Wang–Berger line are not invoked.

### *S1.4 Naming*

The exact phrase *metric random tiling* returned zero hits in a September 2026 check that ran four parallel search lines (classical random tiling; mathematical existence and enumeration; soft-matter neighbours; materials and naming) over arXiv, CrossRef, OpenAlex, and web search (≈ 70 query groups), with no instance of the statistical-mechanics layer studied here (equiangular tilings whose

only disorder is metric, with counted entropy, diffraction, or phason-type elasticity); stated confidence is moderate-to-high, with two documented blind spots (pre-1990s literature outside CrossRef/arXiv, and in-progress 2026 preprints), and a Scopus/WoS re-verification with archived query strings is a submission-stage item. Theory-lineage neighbours are separated by operationally measurable criteria in Table S1.

### *S1.5 Boundary statements*

*Anisotropy boundary.* The displacement-spectrum ratio $S\perp/S\parallel = 0.14–0.17$ is a measured property of the metric field's fluctuation spectrum (S10.3); a strain theory that would turn it into elastic constants of a discrete-metric tiling does not yet exist.

## S2. Classification catalogue: the eleven-network table

Table S2 collects the full classification. Layer discipline: the $\mathbb{R}$ column is theorem-level (dimension of the real solution space); the $\mathbb{Z}$/count column and $s_\infty$ are exact counts or numerical determinations with stated calibre. For the honeycomb, the $\mathbb{Z}$ layer is further sorted by alphabet shape class (Sect. S6); the diffraction ensembles of Sect. 2.3 are drawn from the pure-metric value domains themselves (Sect. S9); no separate counting layer is involved. The Penrose rhombus family is settled by the strip theorem (Sect. S3) and is not a vertex-type network, hence not a row here.

## S3. Propagation mechanisms: square lattice, triangular lattice, and the strip theorem

**Theorem S3.1 (zero metric entropy on the square lattice) [T].** Let G be the Lx×Ly square-lattice graph on a torus, every face required to be a rectangle (all angles 90°) and every edge length drawn from a finite alphabet of size k. Then $\Omega = k^{Lx+Ly}$ exactly, and $s = (Lx+Ly) \ln k /(Lx\, Ly) \to 0$.

*Proof.* The rectangle condition forces top = bottom and left = right on every face; chaining in j equates all horizontal edges of a column to one length, and symmetrically for rows. Conversely any $(h_1 \ldots h_{Lx}; v_1 \ldots v_{Ly}) \in A^{Lx+Ly}$ defines coordinates by cumulative sums and closes periodically. The residual O(Lx+Ly) row/column modes are sub-extensive. (Scope: the square-grid combinatorics; in a general quadrangulation the propagation classes are straight-ahead paths and need not be sub-extensive.) ■

**Theorem S3.2 (rigidity of the triangular lattice) [T].** In any connected 6-valent triangulation with all angles 60° and edge lengths in a finite alphabet, all edges share one common length: $\Omega = k$ on the torus, $s = 0$.

*Proof.* Every triangle is equilateral; equilaterality propagates across the connected triangle-adjacency graph. The constraints are overdetermined (2F = 4V independent equalities against E = 3V variables). ■

**Theorem S3.3 (strip theorem; Penrose family) [T].** In any edge-to-edge tiling whose faces are equiangular parallelograms (rhombi) with edge lengths from a finite alphabet of size k, opposite sides of every face are equal, and this pairing propagates from edge to edge through faces; connected components ("strands") carry constant values, and every strand-value assignment closes every face: $\Omega = k^{\#strands}$ exactly. For Penrose tilings, de Bruijn's pentagrid duality[S4] identifies strands with grid lines, of which a window of radius R meets O(R), while the number of faces is $O(R^2)$: $s = 0$. The square lattice result is the 90° special case. Note the average valence of any rhombic tiling is $\bar{z} = 4$: the zero-entropy mechanism is propagation, not overdetermination.

*Numerical validation.* $\Omega = k^{Lx+Ly}$ verified exactly on test tori; strip counts $\Omega = k^{\#strands}$ verified on generated rhombic tilings (`SQ_数值验证.py`, `TRI_数值验证.py`, `PEN_数值验证.py` + CSVs). Figure S2a illustrates the propagation mechanism, Fig. S2b the strand structure on a rhombic patch (drawn periodic for legibility; for Penrose tilings the strands are the Ammann worms of the de Bruijn grid).

## S4. Spectral mechanism: kagome

**Theorem S4.1 (zero metric entropy on the kagome lattice) [T].** Let G be the Lq×Lr kagome graph on a torus (the medial graph of the honeycomb; faces equilateral triangles and hexagons in ratio 2:1, angles 60°/120°), edge lengths from a finite alphabet of size k. Then $\Omega \le k^{(Lq+Lr+\gcd(Lq,Lr)-1)}$, and $s = 0$ (zero-line structure in Fig. S2c).

*Proof.* Each triangle forces its three lengths equal (one variable per honeycomb vertex); each hexagon imposes two independent pair-sum equations, giving a 2F×2F real linear system M. M commutes with torus translations, so $\ker M = \bigoplus_k \ker \hat{M}(k)$ with 2×2 symbol $\hat{M}(a,b)$, $a = e^{ik_1}$, $b = e^{ik_2}$:

$$\hat{M}(a,b) = \begin{bmatrix} -1 - a^{-1}b + 2a^{-1} & -2 + a^{-1} + b^{-1} \\ 2 - a^{-1}b - a^{-1} & 1 - 2a^{-1} + b^{-1} \end{bmatrix}$$

whose determinant factors exactly (clearing $a^2b$):

$$\det \hat{M} = 3(a-1)(b-1)(a-b) / a^2b.$$

(The symbol entries are Laurent polynomials in a, b; the factor $1/a^2b$ is their cleared common denominator and is nonzero everywhere on the Brillouin zone, so the zero set of $\det \hat{M}$ is exactly the zero set of the cubic polynomial $3(a-1)(b-1)(a-b)$.)

The zero set is three wrapped lines $\{k_1=0\}$, $\{k_2=0\}$, $\{k_1=k_2\}$; the momentum grid meets it in $Lq+Lr+\gcd(Lq,Lr)-2$ points, with nullity 2 at the origin (uniform + staggered modes) and 1 elsewhere, so $\dim \ker M = Lq+Lr+\gcd(Lq,Lr)-1$. Gaussian elimination parametrizes the solutions: $\Omega \le k^{\dim \ker M}$. The theorem is one-sided: it pins $s = 0$ without determining Ω itself (unlike the counting closed forms of Sect. S3/S6). ■

The residual modes are three stripe families (row-, column- and diagonal-modulated with staggered sign). Mechanistically the kagome branch is *spectral* (isolated zero lines of a non-vanishing symbol determinant), against the *propagational* square lattice (chained equalities); the honeycomb's symbol is rank-deficient everywhere (extensive freedom), and the triangular lattice is overdetermined. The kagome contains hexagons but pins them with equilateral triangles, and it is the counterexample that completes the "one hexagon, one module" criterion of the main text.

*Numerical validation.* dim ker = Lq+Lr+gcd−1 verified bit-exactly (Fraction arithmetic) and $\Omega \le k^{dim}$ on test tori (`KAG_numerical_verification.py` → `KAG_entropy_density_verification.csv`).

## S5. Split rows: 4.8.8 on $\mathbb{Z}$ and on $\mathbb{Q}(\sqrt{2})$

### *S5.1 $\mathbb{Z}$ library: discretization-induced rigidity [T]*

**Theorem S5.1.** On the L×L truncated-square (4.8.8) torus with equiangular rectangles and octagons and edge lengths in $A \subset \mathbb{Z}$, $\Omega = k^{4L}$ exactly (class number 4L), $s \to 0$, although the real solution space has extensive dimension $dim_{\mathbb{R}} = 2L^2+2$ (strand structure in Fig. S2d).

*Proof.* The octagon closure projections mix coefficients 1 and √2/2. For integer (rational) edge values, each equation $\Sigma(a_i+b_i\sqrt{2})s_i = 0$ splits into the pair $\Sigma a_i s_i = 0$, $\Sigma b_i s_i = 0$ (rational independence). The split octagon rows combine into four opposite-side equalities; squares contribute two. All constraints are equalities; opposite-pairing propagates along the four direction families into strands, and $\Omega = k^{\#strands} = k^{4L}$, while the un-split real system retains one free variable per face. ■

3.12.12 (√3/2 splitting) does *not* fall to the same mechanism: the split rows partition its constraints into a triangle-locked cut sector (sub-extensive) and a mid-edge sector that inherits the honeycomb's extensive deficiency, with $dim_{\mathbb{Q}} = L^2+3L+1$ (quadratic, exact L = 1…6) and a positive count at every alphabet width (S5.3). The exclusion therefore extends only across the mixed-face z = 3 family wherever the split rows exhaust the constraints, that is, wherever every constraint becomes an equality and the split-solution lattice is the strand lattice itself (4.8.8: 4L strands). Where the split-solution lattice stays *extensive*, as in the 4.6.12 case (split rank $4L^2+4$, S8.2), the $\mathbb{Z}$ verdict reverts to a genuine counting question, answered by the transfer-matrix spectral determination (S8.2): the perimeter-law route to zero entropy is falsified at strict level, and the network is strictly positive at every alphabet width accessed rather than excluded. The four canonical graphs (60°/90° direction algebra) are split-free, which is precisely why their classification survives discretization.

### *S5.2 $\mathbb{Q}(\sqrt{2})$ lattice library: dimension revives, the count stays sub-extensive [N]*

On the lattice $\Lambda = \mathbb{Z} + (\sqrt{2}/2)\mathbb{Z}$ the dimension doubles exactly ($dim_\Lambda = 2\, dim_{\mathbb{R}}$). The library is the ternary non-arithmetic base $\{1, \sqrt{2}/2, 1 + \sqrt{2}/2\} \subset \Lambda$; the integer library {1, 2, 3} realises the $\mathbb{Z}$ baseline $3^{4W}$ against which the excess g(W) below is measured. (The Λ base admits no calibrated tree-sampling estimate: randomised search on these strand-constrained systems finds no live branch at W

≥ 3, so the counts rest on the exact and stochastic-trace engines below.) Its counting entropy was adjudicated by exact/controlled counting on W×W tori (F = 2W²; table below, CSV `TSQ_Lambda_DP_exp.csv`):

| W | F | Ω | ln Ω | ln Ω/F | $g(W) = \Omega/3^{4W}$ | Δs |
|---|---|---|---|---|---|---|
| 2 | 8 | 6 641 | 8.8010 | 1.1001 | 1.0122 | 0.0015 |
| 3 | 18 | 563 553 | 13.2420 | 0.7357 | 1.0604 | 0.0033 |
| 4 | 32 | 52 864 817 | 17.7832 | 0.5557 | 1.2281 | 0.0064 |
| 5 | 50 | 5 291 578 368 ± $1.1\times10^5$ | 22.3894 | 0.4478 | 1.5176 | 0.0083 |

W = 2, 3, 4 are exact (DP = DFS; W = 4 by block-trace = $\Sigma(T^2 \circ T^{2T})$ double method); W = 5 is a Hutchinson stochastic trace[S5] (20 000 probes × 8 seeds, ±0.0022%) on a matrix with 62.2M nonzeros. The excess over the $\mathbb{Z}$ baseline stays small and grows slowly (Δs = 0.0015 → 0.0083 across W = 2…5, its increments already non-monotonic: 0.0018/0.0031/0.0019, Fig. S3), consistent with an approach to a microcanonical entropy platform δ* ≈ 0.008–0.010. Independently, the count series itself grows linearly in W: ln Ω/W = 4.40, 4.41, 4.45, 4.48 across W = 2…5 (linear fit slope 4.53), i.e. ln Ω ≈ 4.5 W + O(1), linear and not area, so the count is sub-extensive (O(W) against F = 2W²) and $\boldsymbol{s_A = 0}$ **at the numerical level**; the verdict is provisional at this calibre.

*S5.3 3.12.12: the mid-edge carrier [N]*

The √3/2 row splitting partitions the constraints exactly. Exact Fraction-rank elimination on L×L tori (L = 1…6; three-way cross-validation as in S11) gives $\boldsymbol{dim_{\mathbb{Q}} = L^2+3L+1}$ (5/11/19/29/41/55 at L = 1…6; second difference constant at 2, quadratic, correcting the linear record superseded from S5.1). After splitting, the two edge subsystems decouple with no shared variables: the **cut sector** (truncated-edge subsystem, $6L^2$ edges in the 60° direction net) is triangle-locked (every triangle forces its three split values equal), leaving $dim_{cut} = 3L-1$ (linear, sub-extensive); the **mid sector** (mid-edge subsystem, $3L^2$ edges) carries the extensivity, $dim_{mid} = L^2+2$, and the identity $dim_{\mathbb{Q}} = dim_{cut} + dim_{mid}$ holds at all six sizes. The entropy question is thus a counting question on the mid sector, answered by the cylinder transfer matrix (the spectral-radius engine of S8.2/S11).

*Transfer-matrix spectra.* The mid-sector cylinder operator's Perron root $\rho(W) = e^{\lambda(W)}$ is the exact L→∞ per-column growth rate (no finite-length transients); the engine passes the gate discipline of S11, with machine-bitwise Tr-vs-DP anchors (64/256/1024 at k = 2, W = 2; 1017/16 041 at k = 3, W = 2). The λ(W) series grows near-linearly in W:

| W | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| k = 2 | (0.897)* | 0.967 | 1.212* | 1.325 | 1.575* | 1.736 | 1.990* |
| k = 3 | 1.609 | 2.012 | 2.502 | 3.015 | 3.549 | 4.093 | 4.643 |
| k = 4 | 2.140 | 2.765 | 3.542 | 4.306 | 5.119 | — | — |

(* = ± conjugate pair at even W, modulus quoted. The k = 2, W = 2 row is excluded from the fits: its spectral top √6 is a cyclic block whose trace contribution cancels for every L (the smallest instance of the selection rule of S6.3), while the k = 3, W = 2 value (ρ = 5) is physical, confirmed by non-commensurate DP chains climbing toward it out to L = 57.)

The per-face entropies s(W) = λ(W)/(3W) converge from above (k = 3: 0.268/0.224/0.209/0.201/0.197/0.195 at W = 2…7). Fitting the symmetric form of S6.3 with W = 2 excluded gives $\mathbf{s_2 \approx 0.13}$ **per face** (window band 0.07–0.13 across fit forms), $\mathbf{s_3 \approx 0.186}$ **per face** (seven widths, rms 0.0007), and $\mathbf{s_4 \approx 0.26–0.37}$ **per face**; the nesting $s_2 < s_3 < s_4$ holds in every calibre. **Verdict: 3.12.12 is a positive-entropy carrier at every alphabet width**. The split rows kill the cut sector but leave the mid sector, which is geometrically the honeycomb's own edge system (below). At the ternary alphabet the positivity is constructive: a twelve-support-class patch certificate gives s ≥ (ln 2)/18 per face at every system size (Theorem S8.3.3).

*Transient exact sub-law sector.* Short-chain DP counts sit on an exact closed form, $\Omega_{mid} = 2^{L+W+1}$ (17/17 non-resonant points at k = 2, bitwise). The transfer-matrix spectrum identifies this as a transient: the sub-leading eigenvalue 2 carries a huge degeneracy whose crossover scale L*(W) grows rapidly with W (k = 3, W = 2: still climbing at L = 57). The same mechanism, with sub-law value ln k, operates on the honeycomb, where the crossover points are measured directly (S6.3).

*Spectrum equivalence with the honeycomb.* The mid-edge subsystem of the truncated honeycomb is the honeycomb's edge system itself, and the spectra agree accordingly: binary counts of the 3.12.12-mid system and the binary honeycomb agree bitwise at eight test sizes, and the ternary-arithmetic λ(W) series agree to $\leq 3\times10^{-4}$ across W = 2…8. The finite-size counting families nevertheless differ (the honeycomb 3×3 count 22 767 against the mid system's 2×3 count 1,017). The transient sectors, not the asymptotes, separate the two systems: same universal width response, different microscopic state space. Data: `TSD31212_tm_*.csv`, `TSD31212_symmetric_fit.csv`, `TSD31212_tm_deep.csv`; scripts `step7_tm.py`, `step7_cross.py`, `step7_deep.py`, `step12/13` (`z3_31212/`).

# S6. Honeycomb geometric limit: affine shape classes, spectral width scaling, resonance

*S6.1 Affine invariance [T]*

The closure conditions are homogeneous and linear, so the solution set over $\mathbb{R}$ is invariant under affine maps of the alphabet (translation x → x+c, dilation x → mx, reflection x → c−x), and the entropy density depends only on the affine *shape class* of A. The ternary phase diagram therefore has exactly three representatives, labelled {1,2}, {1,2,3}, {1,2,4} throughout (all counts are invariant under integer translation of the alphabet, S7.3 Lemma 2).

*S6.2 Width scaling: transfer-matrix spectra and the mixed global model*

The per-column growth rates are the cylinder transfer-matrix spectral radii $\lambda(L_y) = \ln \rho(L_y)$, exact asymptotic objects (Perron roots, no finite-length transients), computed on explicit sparse operators with ARPACK (S11). Validations: $\mathrm{Tr}(T^L)$ reproduces the exact DP counts bitwise at anchor sizes for every class; at even widths the leading eigenvalues form ± conjugate pairs or roots-of-unity families, so the quoted λ is the spectral modulus, cross-checked by dense diagonalisation ($L_y \leq 6$) and trace fits (S6.3). Results (Table S4: the quoted values are the spectral moduli of `TSDHC_binary_spectral.csv` and `TSDHC_nonarith_spectral.csv` together with the odd-width rows of `TSDHC_tm_rho.csv` and `TSDHC_tm_rho_nonarith.csv`; the even-width rows of the two tm_rho files are unconverged power-iteration prefixes, superseded by the spectral files):

$\lambda(L_y)$: binary 0.89588/0.96657/1.24245/1.32452/1.59197/1.73615/1.99572/2.17424 ($L_y$ = 2…9); non-arithmetic 1.24245/1.38629/1.50444/1.58305/1.78945/1.89771 ($L_y$ = 2…7); arithmetic 1.60944/2.01215/2.50211/3.01488/3.54934/4.09316/4.643 ($L_y$ = 2…8; the $L_y$ = 8 value is the 3.12.12 interlocking check, S5.3).

*Mixed global model.* The per-tile values $\lambda/L_y$ converge exponentially in $L_y$ (consecutive increments decay geometrically), so neither single-form extrapolation (pure $c/L_y$ or pure geometric remainder) captures both limits. The mixed form $\lambda/L_y = s + c/L_y + A'q^{Ly}$ fits all measured widths of each class simultaneously (binary: even/odd branch lines sharing s, c, q, with A′_odd = 0.35 A′_even):

| class | s∞ | c | A′ | q | rms |
|---|---|---|---|---|---|
| arithmetic {1,2,3} | **0.544** | 0.289 | 1.607 | 0.270 | 0.0007 |
| non-arithmetic {1,2,4} | 0.224 (free fit) | 0.221 | 0.870 | 0.574 | 0.0038 |
| binary {1,2} | 0.209 (amplitude ratio pinned) | 0.268 | 0.320 (even) / 0.112 (odd) | 0.572 | 0.0024 |

*What the width series resolves.* The arithmetic row is pinned by the data: every legal form (pure $c/L_y$, geometric remainder, free amplitude ratios) places its asymptote within 0.53–0.58, and s = 0.544 is the free fit. The other two rows are not pinned: at six and eight points the four-parameter form is under-determined. For the non-arithmetic class, s anywhere in 0.17–0.27 changes the rms by less than the fit's own scatter (profile of the nine-point series: rms/rmss* ≤ 1.08 for s ∈ [0.17, 0.27]; rms = 0.003781 at s = 0.209, 0.003752 at the free-fit 0.224, 0.003794 at 0.242; the three values agree to three digits). For the binary class, releasing the even/odd amplitude ratio from the pinned 0.35 to its free value moves the fit to s = 0.248 (rms 0.00125 against 0.00239); the three legal amplitude ratios (0.35 pinned, 1.41 free-optimal, 2.67 at the pinned-s optimum) disagree, and the asymptote follows the pin. The width series therefore separates the arithmetic class from the other two, and places the binary and non-arithmetic classes together near 0.2 inside the common band 0.17–0.27, a band wider than their difference. What the spectra settle with no fit at all is the per-width nesting $\rho_{013}(W) > \rho_{01}(W)$, which holds at every measured width. Both pinned asymptotes are anchored from below

constructively, $s \geq (\ln 2)/9$ (arithmetic, every size) and $s \geq (2/25) \ln 2$ (binary, every width; $(\ln 2)/4$ on even widths) (Theorems S8.3.2, S8.3.6); the non-arithmetic class has no constructive bound yet. Superseded readings of the earlier draft (arithmetic 0.45/0.53, binary 0.18 (c/W) and 0.25 (limsup misread), non-arithmetic 0.13 (c/W) and 0.290 (three-point geometric)) are withdrawn to the repository audit trail.

*Tile-library enumeration.* The libraries behind Fig. 3b are enumerated by exact closure: on {1,2,3} there are exactly 45 closed hexagons, falling into 15 dihedral orbits (sizes 1,3,2,3,3,6,6,2,1,6,3,3,2,3,1; Fig. S4); on {1,2,4}, 33 closed hexagons in 13 orbits.

*S6.3 Resonant and non-resonant branches: the symmetric form and the selection rule (main text Fig. 3d,e; SI Fig. S5)*

*Branch structure.* For every class and width the counts split by the transposition phase $L_x$ mod $L_y$. On non-resonant coprime sizes ($L_x \not\equiv 0 \bmod L_y$ with $\gcd(L_x, L_y) = 1$), the binary and non-arithmetic counts follow the perimeter law $\ln \Omega = \ln k \cdot (L_x + L_y + 1)$ **bitwise**: machine-zero deviation at every such measured size, short ($L_x \leq 20$) and long ($L_x \leq 150$) alike. On commensurate sizes ($L_x \equiv 0 \bmod L_y$) the count exceeds the closed form by an amount that grows approximately linearly in $L_x$ at fixed $L_y$ (local exponent 0.96–1.10; Fig. S5b; up to $7.7 \times 10^4 \times$ the closed form for binary at 20×5): these are the winding sectors, and the transfer-matrix spectra of S6.2 show that their asymptotes, not the sub-law, carry the extensive entropy. The arithmetic class crosses over to its asymptote within 10–20 columns and never sits on the closed form (the ternary 3×2 count is 1017 against the binary 64 at the same size).

*Symmetric form.* Transposition symmetry $\Omega(L_x \times L_y) = \Omega(L_y \times L_x)$ forces the counting form to be symmetric under $L_x \leftrightarrow L_y$; the mixed model of S6.2 is its $\lambda(L_y)$ limit, with the symmetric generalisation

$$\ln \Omega(L_x, L_y) = a \cdot L_x L_y + b(L_x + L_y) + A \cdot L_x L_y (q^{L_x} + q^{L_y})/2,$$

whose three terms are the area term (the extensive entropy), the perimeter term (the perimeter law is the $a = 0$ special case), and the two-sided width-modulation term: positive for the arithmetic and binary classes (channel-opening excess), negative for the non-arithmetic class (sub-opening deficit).

*Selection rule (why the sub-law looks exact).* Beyond the sub-law sector the transfer-matrix spectrum organises the extra sectors as W-th roots-of-unity families, whose trace contribution vanishes exactly whenever $W \nmid L$; at even widths the spectrum carries in addition ± pairs (the same families doubled in a different momentum sector), whose contribution $\rho^L(1 + (-1)^L)$ vanishes only at odd L. The exact-size rule, however, is the coprimality condition $\mathbf{gcd(L_x, L_y) = 1}$ for the sub-law classes, stated with its full form and verification at the end of this section: the vanishing conditions above are necessary but not sufficient, and the archived long-chain counts show blocks that satisfy $W \nmid L$ yet violate the

closed form — at the even width $L_y = 4$, the non-resonant sizes with $L_x \equiv 2 \pmod 4$ exceed it (24 of 71 rows; the deviation grows from 8× at $L_x = 10$ to $4.6 \times 10^{12}$ at $L_x = 98$) — while every measured coprime block (binary and non-arithmetic, $L_y = 2, 3, 5, 514$ rows together) follows it bitwise. The anomalous block still rides the width-4 asymptote: $\ln \Omega/L_x$ falls monotonically to 1.568 at $L_x = 98$ against $\lambda(4) = 1.504$, so $\lambda(4)$ is unaffected, and the widths of Fig. 3d ($L_y = 2, 3, 5$) are clean. Direct observations match: the binary W = 5 sub-law breaks between L = 21 and L = 25 (the first large 5|L size; the count overshoots $2^{Lx+Ly+1}$ by a factor 1,680), W = 3 breaks between L = 11 and 13; at (W, L) = (6, 11) the count is still exactly $2^{18.}$ The perimeter law is thus a precise statement of entropy inaccessibility at the exact sizes. Two-stage convergence: the apparent per-column slope departs from ln k and converges to $\lambda(W)$; the crossover scale is ~10–20 columns (arithmetic), hundreds (binary), and ≥ thousands (non-arithmetic).

*Fixed-aspect-ratio families.* Every family $L_x = m \cdot L_y$ with integer m is commensurate, and so is every square torus ($L_x = L_y$). Along the binary family m = 4 (exact counts; 8×2, 12×3, 20×5 from the archived series; 28×7 and 36×9 computed for this section):

| $L_x \times L_y$ | Ω | per-tile ln Ω/F | closed form | excess per tile |
|---|---|---|---|---|
| 8×2 | 5 696 | 0.5405 | 0.4766 | 0.0639 |
| 12×3 | 1 166 278 | 0.3880 | 0.3080 | 0.0800 |
| 20×5 | 5 168 846 812 714 | 0.2927 | 0.1802 | 0.1127 |
| 28×7 | 30 745 494 532 694 531 373 190 | 0.2642 | 0.1273 | 0.1369 |
| 36×9 | 308 008 477 964 244 299 420 516 558 126 679 400 | 0.2522 | 0.0984 | 0.1538 |

The excess per tile rises monotonically with width (0.064 → 0.154 across $L_y = 2 \ldots 9$). The square family confirms the same behaviour at m = 1 (the standard thermodynamic-limit shape): 9×9 gives Ω = 22 285 903 280 (per tile 0.2942; excess 0.1316, against 0.1199 at 7×7). And at $L_y = 9$ the excess per tile rises with aspect ratio across the board, 0.1316 (m = 1), 0.1448 (m = 2), 0.1506 (m = 3), 0.1538 (m = 4), tracking the asymptotes $\lambda(L_y)/L_y$ of S6.2: **the commensurate families carry the winding-sector entropy itself, rising at every computed width toward the class's s∞, not an add-on to a sub-extensive base** (CSV `P0_PM_resonant_families_Ly9.csv`; parallel counting anchored at 3×3/8×2/7×7/9×9, S11).

*Dimension does not jump.* Fraction-exact rank elimination of the closure system at resonant and non-resonant sizes alike (2×2, 3×3, 4×2, 5×2, 4×3, 7×5, 7×7, 8×7, 10×5, 14×7) gives $dim_{\mathbb{R}} = F + 2$ at every size: the excess is not a dimension change but a $\mathbb{Z}$-thinness effect: the same real solution space is populated more thickly by integer points when the gluing commensurates. The entropy question is thus arithmetic on the side of the counting domain as well as of the alphabet: it reads the arithmetic of the sequence, not only of the values.

*Consequence for the classification (superseded verdicts).* The two zero-entropy verdicts that Table S2 carried until this revision (binary and non-arithmetic "s = 0 on generic sequences") are withdrawn (2026-10-07): the exact perimeter law on which they rested is the winding-off branch (selection rule above), and the transfer-matrix spectra show every class positive (S6.2). The commensurate-sector readings of this section are the same entropy at fixed aspect ratio, not an add-on. Whether the planar open-boundary count realises the winding-sector value is treated in the next paragraph.

*The planar open-boundary count [N].* The open question recorded above is now answered by direct counting on simply connected patches. The exact-size rule is a conjunction of two arithmetic conditions: the closed form $\Omega = k^{Lx+Ly+1}$ holds bitwise precisely when gcd($L_x$, $L_y$) = 1 **and** the alphabet contains no three-term arithmetic progression (no a < b < c in the alphabet with a + c = 2b), and is exceeded otherwise (e.g. at ($L_x$, $L_y$) = (9, 6), gcd 3; the arithmetic alphabet, which carries the progression 1 + 3 = 2·2, exceeds the closed form already at coprime sizes). It is verified bitwise on the delivered grid of 2,232 counting rows (2,088 distinct alphabet–size points; 30 alphabets, k = 2–8, $L_y$ = 2–6, $L_x$ up to 150) with zero violations, re-deriving the prediction per row from the two conditions; it subsumes the $L_y$ = 4 block above as a consistency check, and stands as a numerically established criterion, with the analytical proof open (grid and criterion documents: Table R2). For a hexagonal-envelope disk of R rings (N = 3R(R+1)+1 faces, 6(2R+1) boundary edges, no periodic identification) the exact counts at R = 3–7 follow ln Ω_disk = a·N + b·R + c with a = 0.2308 ± 0.0005, b = 3.988 ± 0.02 and c = 2.22 (rms 0.001; every sub-window of R = 3–7 gives a in 0.2303–0.2313, and the local second-difference slope converges from 0.221 to 0.231; adding one small geometric correction term moves the area coefficient to 0.2320–0.2325, in agreement with the independent geometric extrapolation of the local slope, 0.2321). The binary-class planar bulk entropy density is positive, a = 0.231(1) nats per face (a/ln k = 0.333(1)), and lies inside the systematic band of the cylinder mixed model above (0.209 pinned / 0.248 free); the same engine at k = 3 gives a = 0.560 ± 0.004 (R = 1–4), inside the arithmetic-class fit band 0.53–0.58. These values carry no width extrapolation and reproduce the class-resolved densities of this section. The boundary enters the planar count with its own measurable configurational entropy, b = 3.99(2) nats per ring (0.332 per boundary edge), alongside the corner term c. The counts are obtained by a column-recursion frontier DP in dense 128-bit accumulators (R = 3–7 for the binary, R = 1–4 for the arithmetic class), independently cross-checked by a directional twin implementation (scatter and gather modes) and, at small R, by full and blocked DFS enumeration; all torus-side conventions are anchored bitwise against the enumeration engines (Reproducibility Supplement, Tables R1/R2).

*S6.4 Corroborating strip counts, and caveats*

Narrow cylindrical strips (W = 2, 3; L = 1…18) reproduce the same ordering through the strip-direction flux λ(W=3): binary 2.55 < non-arithmetic 3.55 < arithmetic 6.95 (pre-crossover finite-window readings, retained as corroboration of the ordering only; the asymptotic values are the spectra

of S6.2), with DFS/DP/brute-force three-path agreement on all 34 test cases (`P0_PM_strip_growth.csv`). Caveats recorded for completeness: (i) W = 1 strips are degenerate (three-hole vertex keys) and are excluded; (ii) the affine fundamental domain used for counting and the standard-embedding torus drawn in Fig. 3a are *different shear classes*, non-isomorphic at finite size (binary 3×3: Ω = 518 vs 842 at the same F = 9); all counts are on the fixed affine convention (S1.2), and no shear-class equivalence beyond that convention is asserted here; (iii) torus identifications merge parallel edges if edge keys are built from hole pairs only; all production counters key edges by vertex triples, which avoids the collapse.

*S6.5 $\psi_6 \equiv 1$ is an identity [T]*

On the honeycomb every vertex joins three edges at exactly 120°, so every nearest-neighbour bond direction is one of three lattice axes differing by 60°. The bond-orientational order parameter $\psi_6^{S6} = \langle e^{6i\theta} \rangle$ therefore equals 1 bond-by-bond, for *any* assignment that respects closure, at any entropy level: bond-orientational order is never degraded by metric disorder. This is the analytic pillar behind the diffraction statements of Sect. S10.

## S7. 3.4.6.4: triangle locking and the area law

*S7.1 Triangle locking [T]*

In any solution on the 3.4.6.4 (rhombitrihexagonal) network: (i) every triangle's three edges are equal (closure with 60° angles and independent directions); (ii) every square forces the two triangle-flanking edges (which are opposite sides) to be equal; (iii) the triangle-adjacency-through-squares graph is the honeycomb graph, which is connected. Hence **all triangle edges in any solution carry one global value t**, and the $6L^2$ triangle edges (half of all edges) contribute a single degree of freedom: the constraint rows have kernel projection rank one on the triangle sector. Together with the square pairings this is the integer-rigidity mechanism that compresses the $\mathbb{R}$-extensive solution space ($dim_{\mathbb{R}} = L^2+3$, theorem) to a leak.

*S7.2 Exact counts and the area law [N]*

Exact torus counts by sparse transfer-matrix trace (binary integer alphabet {1,2}; integer arithmetic; Table S5, CSV `RTH_torus_area_law_8pts.csv` + trace logs):

The s·L series is U-shaped (Fig. S6; transient line dominates at small L, area term accelerates from L ≥ 6). The fit windows bracket the asymptotic slope: equal-weight quadratic over all nine exact points gives $\ln \Omega = 0.733\ L + 0.186\ L^2 + 2.91$ (RMSE 0.18), $s_\infty = b/6 = 0.031$; the asymptotic window L = 5–10 gives $b \approx 0.244$, $s_\infty =$ **0.040** with RMSE 0.0028, a factor of 65 below the nine-point protocol. The two registered out-of-sample predictions were hit by the asymptotic-window form: $\ln \Omega(9) = 24.520$ landed on the registered asymptote, and $\ln \Omega(10) = 29.018$ against the registered 29.03 (deviation 0.012; the alternative global line, whose L = 6…9 linear extrapolation gives 27.81, misses

the measured point by ≈ 1.2 in ln Ω and is excluded). Protocol locked for the manuscript: we quote **$s_\infty$ = 0.040** (asymptotic window) and report 0.031, the equal-weight nine-point value, as the fit-protocol systematic (Table S3), so the protocol span is 0.031–0.041. A three-parameter cubic on the asymptotic window has a negative leading coefficient ($-1.4 \times 10^{-3}$): the form itself is unstable at these window lengths, and the quadratic is the registered form. The qualitative conclusion, positive entropy density a factor of 5–7 below the honeycomb's thinnest alphabet class (0.21, S6.2), is protocol-independent, and by the alphabet monotonicity of S7.3 it is a lower bound for every alphabet containing two distinct values, the chemically motivated {4,5,6} included.

*S7.3 Alphabet monotonicity: the binary count bounds every richer alphabet*

The nine-point series of Table S5 is the binary integer alphabet {1,2} (translation-equivalent to {4,5}, a two-value subset of the chemical alphabet {4,5,6}). Three elementary lemmas fix its scope.

**Lemma 1 (inclusion monotonicity).** If $A \subseteq A'$ then $A^E \subseteq A'^E$ and the closure constraints are unchanged, so $\Omega_A(L) \leq \Omega_{A'}(L)$ at every L; wherever both entropy densities converge, $s_\infty(A') \geq s_\infty(A)$.

**Lemma 2 (translation invariance).** On every closed face the directed edge vectors sum to zero, so $x \to x + c \cdot 1$ maps solutions to solutions for any integer c: counts are translation-invariant in the alphabet, $\{a, a+d\} \cong \{1, 1+d\}$ (numerically: shift-start variants, alphabet offset 0 and 1, give identical counts in the transfer-matrix pipeline).

**Lemma 3 (homogeneity).** The face-closure constraints are homogeneous and linear, so the solution set S is a linear subspace; combined with Lemma 2, scaling about the shift $x \to 1 + d \cdot (x-1)$ is a bijection between $S \cap \{1,2\}^E$ and $S \cap \{1,1+d\}^E$. Hence **every two-value alphabet {a,b} counts the same as {1,2}, at every L, exactly**: no "consecutive values" restriction is needed.

**Corollary.** Every alphabet containing two distinct integers (in particular the chemically motivated {4,5,6} and the non-consecutive {4,6}) satisfies $\Omega \geq \Omega_{binary}$ pointwise, so its entropy density, where it converges, obeys $s_\infty \geq 0.031$ (the conservative protocol of the binary determination, S7.2). The ternary counts already in hand obey the bound at every computed size: $\Omega(\{1,2,3\}) = 2187 / 68\,301 / 3\,837\,303$ against the binary 128 / 1 036 / 8 800 at L = 2/3/4 (README `z4_3464`). The corollary is a lower bound only; it does not evaluate $s_\infty$ for richer alphabets. The ternary positivity is now a constructive theorem (a 19-face patch certificate, $s \geq (\ln 2)/24$ per face at every size; Theorem S8.3.3); the monotonicity lemmas keep their role on the binary side and for every richer alphabet.

**Matched-width contrast.** On the honeycomb the same lemmas hold with the shape-class classification (Sect. S6): every two-value alphabet is affine-equivalent to {1,2}, whose winding sectors carry s ≈ 0.2 per tile (S6.2); the exact perimeter law on non-resonant sizes is the winding-off branch, not a zero-entropy verdict (S6.3). At matched binary width the ordering reads: honeycomb ≈ 0.2 per tile, 4.6.12 ≥ 0.186 per face at width 3 (S8.2), 3.4.6.4 a leak of 0.040; magnitude, not survival,

is what the closure structure decides at k = 2; the theorem-level zeros (square, kagome, 4.8.8 on $\mathbb{Z}$) are the other side of the same ledger.

## S8. The four-layer ladder, the dead types, and open questions

### *S8.1 Four-layer diagnostic ladder*

L1 *net degrees of freedom* $E-2F$[S7]: necessary only (3.4.6.4 has $E-2F \le 0$ yet positive entropy). L2 *deficiency*: extensive families of constraint-row dependencies localised on single faces: the honeycomb's mechanism, surviving $E-2F \le 0$. L3 *row splitting*: whether the alphabet's number field contains the constraint coefficients' field (Niven dichotomy: the closure-coefficient projection onto its own edge direction is rational $\Leftrightarrow$ the face polygon has $p \in \{3,4,6\}$ sides; only faces outside {3,4,6} can split rows, the sole entry point of alphabet arithmetic). L4 $\mathbb{Z}$*-thinness*: whether integer points of the constraint space are thick (exponential in area) or thin (perimeter-bounded): for finite libraries this, not dimension, is the entropy question. The dichotomy of the main text (finite library = $\mathbb{Z}$-thinness, infinite library = dimension) is L4 vs the $\mathbb{R}$ layer. **L4′ (transient exact sub-law discipline).** A system that admits an exact propagation sub-law (row/column-constant closed-form families) carries it as a sub-leading eigenvalue with a huge degeneracy, and the crossover scale $L$*(W) grows rapidly with W (measured* $L \gtrsim 21$–$57$+). Any width-scaling verdict must therefore rest on a transfer-matrix spectral radius or on demonstrated window-end stability; short-chain $\lambda(W)$ plateaus and ratios are not admissible evidence in this family. Audited: 3.12.12 (struck, positive), binary honeycomb (struck, positive), non-arithmetic honeycomb (struck, hidden carrier), 4.6.12 (struck, perimeter law falsified); exempt: 4.8.8 (exhaustion theorem), 3.4.6.4 (all-rational, no sub-law carrier).

### *S8.2 The eleven uniform types: entropy signs and carriers*

Among the 21 angle-legal uniform vertex types[S8], 10 admit no edge-to-edge tiling at all and hence no metric entropy: eight fall to an odd–even locking lemma (3.7.42, 3.8.24, 3.9.18, 3.10.15, 4.5.20, 5.5.10, 3.4.4.6, 3.4.3.12), two to local chain contradictions (3.3.6.6, 3.3.4.12). The remaining 11 are the Archimedean tilings, and all eleven now have their entropy signs determined (Table S2): the positive carriers are the honeycomb (every alphabet class, S6.2), 3.4.6.4 (S7), and the split-row pair 3.12.12 (S5.3) and 4.6.12 (by the transfer-matrix spectral determination below). Its constraint-matrix layer is settled: exact ranks on L×L tori (L = 2…4; pipeline validated by reproducing the 3.12.12 split dimensions 11/19 digit-for-digit) give $dim_{\mathbb{R}} = F + 2$ (the z = 3 mixed-face family law) and **confirm √3 row splitting**: the split-system dimension is $4L^2 + 4$ (exact at three sizes) against $dim_{\mathbb{R}} = 6L^2 + 2$. Unlike 4.8.8, where the split dimension 4L is sub-extensive and s = 0 follows definitively ($\Omega \le k^{4L}$), the 4.6.12 split-solution lattice retains *extensive* rank, so the $\mathbb{Z}$ verdict is a genuine counting question (S5.1). Exact and validated-stochastic enumeration gives $\boldsymbol{s_{\mathbb{Z}}}$ **= 1.29, 0.717, 0.5465 ± 0.0005, 0.4778 ± 0.0001 at L = 1…4** (Table S6; $\Omega_{\mathbb{Z}}(2\times2) = 29\,723\,517$ exact; $\Omega_{\mathbb{Z}}(4\times4) \approx 8.35 \times 10^{19}$). The Knuth tree-sampling estimator[S9] is anchored before deployment: it reproduces the exact full-orbit totals at the

anchor sizes, 2 346 against 2 349 at L = 1 (30 000 samples) and $2.971 \times 10^7$ against $2.972\,351\,7 \times 10^7$ at L = 2 (48 000 samples; ln deviations −0.0013/−0.0005), with the anchor runs and the estimator version archived in the counts CSV (S11; the naive per-sample s.d. understates heavy-tail uncertainty, as for the Λ lattice). The per-tile entropy decays slowly: ln $s_{\mathbb{Z}}$ versus ln L has local slopes of −0.85/−0.67/−0.47 between consecutive points, too curved for a single power law, and extrapolation frameworks consistent with the four points give 0.16–0.48, with $s_{\mathbb{Z}}(L = 4)$ still an order of magnitude above the 3.4.6.4 leak; but the sequence has not converged, so the ternary series alone carried 4.6.12 as a thick-entropy candidate, a status the transfer-matrix spectral determination below settles (the perimeter-law route to zero entropy is falsified; S8.2), with a mechanism distinct from both the honeycomb deficiency and the 3.4.6.4 locked leak. k-uniform and mixed-vertex generalisations are out of scope.

*At L = 1 the torus is degenerate (equations merge) and is used only for the brute-force anchor; the $4L^2+4$ split dimension is exact at L = 2…4. $\Omega_{\mathbb{Z}}$ counts full translation orbits (gauge-fixed count × 3: the L = 1, 2 exact values are 783 × 3 and 9 907 839 × 3); the L = 3, 4 estimates are quoted directly on this full-orbit calibre, anchored against the exact totals at L = 1, 2. Extrapolations of $s_{\mathbb{Z}}(L)$ consistent with the four points give $s_{\mathbb{Z}}(\infty) \approx 0.16$–$0.48$ (reference readings; the decay has not converged at L = 4). The spectral determination below narrows this window from below by the binary rates (strict readings ≥ 0.186 and ≥ 0.160 per face at W = 3 and W = 4; reference asymptote ≈ 0.09–0.10 per face). Fig. S7a shows the ln $\Omega_{\mathbb{Z}}$–$N_F$ readout with the area/perimeter guides.

**The transfer-matrix spectral determination [N].** The binary system is where the exact ladder reaches furthest, and its verdict decides the sign for every richer alphabet. The closure constraints are pointwise on values and homogeneous linear, so solution sets nest under alphabet inclusion; and by the affine invariance that homogeneous closure confers (the constraint rows sum to zero, A·1 = 0, as everywhere in this paper), the least favourable finite alphabet is any two-value alphabet, equivalent to {1,2} in the main-text labelling, realised as {0,1} in the counting scripts. *Lemma S8.2 (alphabet monotonicity).* If $A \subseteq A'$ are finite integer alphabets, then $\Omega_A(G) \le \Omega_{A'}(G)$ on every counting domain G: every A-solution is an A′-solution. ■

Square-torus anchors: $\boldsymbol{\Omega_{01}}$ **= 112, 22 912 and 11 261 320 at L = 1, 2, 3** (per-tile s = 0.786, 0.418, 0.301), from two independent engines, a prefix-parameterised blocked DFS and a frontier dynamic program over a face-closure ordering (states merged by key; no gauge fixed, full-orbit calibre), agreeing digit-for-digit at every overlap, with brute force at L = 1; the 2×3 torus gives $\Omega_{01}$ = 58 240, likewise dual-engine, and the rectangular 4×3 torus gives $\Omega_{01}$ = 15 300 096 (s = 0.230, depth-split blocked DFS; the transposed 3×4 case runs into the DP's frontier cap). These four values are the machine-bitwise gates (Gate A) of the transfer-matrix construction below.

*Engine.* The verdict is decided by the cylinder transfer-matrix spectral radius, the exact asymptotic per-column growth rate, immune to the finite-length transients (the DP odd-branch flux $\lambda(2) \approx 2.5$ read from the W = 2 series L = 2…14 is a pre-crossover reading; the true rate(2) is 2.747205). The operator is built on per-block local role naming (roles I/O/M/P with covariant ordering): a geometric slot hash is provably impossible here, since any translation-covariant key collides a class with its own column-shifted image. Gates: Gate A, the frontier DP reproduces the four torus anchors bitwise (above); Gate B, the full block product from empty head to empty tail reproduces exact torus counts (58 240 at (2,3), 2 569 312 at (2,4)); the tail-connectivity gate and the known-answer calibration of S11; leading spectra are degenerate (W = 2: $\rho = 15.5990$, five-fold; power iteration fails, ARPACK throughout).

*Verdict: the binary perimeter law is falsified at strict level.* **rate(2) = 2.747205 is pinned two-sided**: the full-head-closure Perron value (an upper bound: the Perron root of a non-negative matrix cannot undercount the true rate) equals the trunk-restricted Perron of a K = 50 true-head sample closure (a lower bound; trunk 100% tail-connectable), with the whole procedure calibrated against this known answer to zero deviation at six digits. **rate(3) ≥ 3.351903 is a strict lower bound**: the trunk-restricted Perron of an independent K = 50 true-head closure (1,324,128 states, trunk 100% tail-connectable, every trapped walk a genuine W = 3 torus configuration); three independent closures (K = 50; K = 300, 8,277,624 states; 5,000 random starts) agree to eight digits. **rate(4) ≥ 3.849459 is a second, independent strict lower bound**, from a single-boundary-state (K = 1) true-head closure (2,840,160 states, 86.4 M nonzeros, trunk 100% tail-connectable; leading $\rho$ six-fold degenerate, the same phase-structure signature as at W = 2 and 3). The strict lower bounds do not rank the widths (rate(3) carries no upper bound): the falsifications at W = 3 and W = 4 are independent, not a monotone chain. Since rate(3) > rate(2) and rate(4) > rate(2), $\lambda(W)$ is not flat; **the zero-entropy perimeter law is falsified (strict, configuration-counting calibre)**, and by Lemma S8.2 the sign transfers to every richer alphabet: **4.6.12 carries strictly positive metric entropy at every accessed width**, with a mechanism distinct from both the honeycomb deficiency and the locked leak: the entropy riding the extensive split lattice. Strict entropy readings: $s_2(W = 2) = 0.2289$ per face; **$s_2(W = 3) \geq 0.18622$ per face**; $s_2(W = 4) \geq 0.160394$ per face — the smaller W = 4 reading measures bound tightness, not a falling $\lambda$. Evidence grade at W = 4: the closure carries no dedicated bitwise Gate B anchor; its strictness rests on the engine's zero-deviation calibration at W = 2 and dual bitwise gates at W = 3, on the constructive containment of every trunk walk in a genuine torus configuration, and on the trunk's 100% tail connectivity; a lost-state defect could only loosen the bound, never inflate it.

*Honest boundary.* Three points falsify the flat law but do not by themselves fix the sign of $s_2(\infty)$: a sub-linearly growing $\lambda(W)$ would still give $s_2(\infty) = 0$. Linear-$\lambda$ slopes of the lower bounds (0.605 per width over W = 2–3, 0.551 over W = 2–4) and a three-point fit $s = s_\infty + c/W$ ($s_\infty = 0.094$, a model-dependent read-out, not a bound: without a decay-form theorem, curves through the three points can

bend the asymptote toward zero) give the consistent reference range $s_2(\infty) \approx 0.09–0.10$ per face (sub-asymptotic indication, not a determination). The ternary side needs no separate computation at sign level: Lemma S8.2 gives $s_3(W) \geq s_2(W)$ at every width, so $s_3(W = 3) \geq 0.18622$ and $s_3(W = 4) \geq 0.160394$ (strict), and the ternary Knuth window 0.16–0.48 stands as a reference reading only. The positivity itself is constructive at the ternary alphabet (Theorem S8.3.3); open are the asymptote and a binary constructive bound. Engine reach and tool boundaries are tabulated in S11.

*Withdrawn width-scaling reading (audit record).* The earlier draft's verdict $s_{01}(\infty) \geq 0.21$ rested on a W = 2…5 strip plateau with "no systematic decay": the same transient-trap reading that struck 3.12.12 and the honeycomb binary class (S8.1 L4′). The DP W = 2 series (L = 2…14) and the calibrated strip-Knuth points (four calibration points at 0.03–2% deviation; weight convention X = Πf) live pre-crossover: their plateau is the sub-law window, not the asymptote, and the perimeter-law control they were read against (the binary honeycomb, "s ∝ 1/W") was itself a transient reading (S6.3). The verdict and its width table are withdrawn; the underlying exact counts remain valid data (Table S6; Fig. S7a). Audit trail (2026-10-07): the initial "two-sided rate(3)" claim and the two-point "$s_2(\infty) > 0$" claim were both downgraded (random-start closures are not upper bounds; two points kill only the flat law), and the tail-connectivity gap was closed by the trunk calculation; the four-gate discipline of S11 records the lesson.

*The three z = 5 rows of Table S2.* The snub square 3.3.4.3.4 has $dim_{\mathbb{R}} = 2$ exact (k = 1…5 verified), the solution space is a two-parameter family, and $\Omega = k^2$ exactly: s = 0 [T] (`z5_snubsquare/`, theorem document with the two-parameter family demonstrated). The 3.3.3.4.4 tiling is sub-extensive, dim = 2L + 1 on the standard tori series: s = 0 [N], with the analytical proof open (`z5_33344/`). The snub hexagonal 3.3.3.3.6 is rigid: $dim_{\mathbb{R}} = dim_{\mathbb{Q}} = 1$ exact at k = 1…5, mechanism triangle-locked propagation (kernel = the complete code ker φ(i,j) = (i + 3j) mod 7 on the triangular lattice), $\Omega \leq k$: s = 0 [T]+[N] (`z5_snubhex/`, theorem document + independent verification passing 5/5).

*S8.3 Constructive lower bounds: local-flip certificates and the binary strip construction*

The positive verdicts of Table S2 are numerical determinations (transfer-matrix spectra and calibrated estimates). This section anchors them constructively. Two mechanisms supply explicit exponential families of configurations: at the ternary alphabet {1,2,3}, every carrier admits a local-flip certificate, a finite patch carrying two admissible completions whose disjoint copies multiply independently; at the binary alphabet, the honeycomb admits a non-local construction built from stripe-compatible site sets. All bounds hold at every system size (ternary) and every strip width (binary): no coprimality or resonance condition, no extrapolation, and certificates that are finite objects under exact machine verification (end of section).

**Lemma S8.3.1 (completion-product bound).** Consider a system of per-face closure equations over a finite alphabet, a legal configuration $T_0$ (the base), and a finite patch P carrying two different legal

completions $\varphi_+ \neq \varphi_-$ that agree with $T_0$ on every boundary edge of P. Then any legal configuration on a region containing n face-disjoint copies of P extends to at least $2^n$ legal configurations, and on the L×L torus $s \geq \rho_P \cdot \ln 2$ per face, with $\rho_P = 1/(d_1 d_2)$ the inverse bounding-box area of P in fundamental-cell units.

*Proof.* The closure constraints are per-face local. Two face-disjoint copies of P share at most boundary edges, on which both completions take the base value, so the choice made in one copy never constrains another; and different choice vectors give different configurations, the flip state being read off the patch interior. ■

**Theorem S8.3.2 (honeycomb, ternary alphabet).** On the honeycomb with alphabet {1,2,3} and base $T_0 \equiv 2$, the seven-face flower patch (a central hexagon with its six neighbours) carries two completions: in $\varphi_+$ the central hexagon takes all edges 1 and every neighbour takes its shared edge 1 and its two radial edges 3; $\varphi_- = 4 - \varphi_+$. Both satisfy the opposite-side sum rules of S1.2 face by face: the centre is equilateral, and each neighbour carries the profile $(v_{k-1}, v_k, v_{k+1}) = (+1, -1, +1)$ on three consecutive edges against zeros elsewhere, which meets both rules. The patch occupies a 3×3 box, so

$\Omega(L) \geq 2^{\lfloor L/3 \rfloor^2}$, hence $s \geq (\ln 2)/9 \approx 0.0770$ per face, at every $L \geq 3$.

*Structure.* The completion difference $v = \varphi_+ - T_0$ is a compactly supported kernel mode with entries in $\{-1, 0, +1\}$ on twelve supporting edges. On width-W windows the space of such modes has dimension exactly $(W-2)^2$, the number of translate positions of a third-difference operator $\Delta_{o0}\Delta_{o1}\Delta_{o2}$, and the translates are linearly independent (each difference is injective on finitely supported fields), so the compact kernel modes of the honeycomb are exactly one third-difference family: the algebraic form of the "one hexagon, one module" criterion. The certificate is drawn in Fig. S8a.

**Theorem S8.3.3 (the three thinner carriers, ternary alphabet).** At the alphabet {1,2,3}, base 2: on 3.4.6.4, a 19-face patch (seven hexagons and twelve squares, no triangle) in a 2×2-cell box gives $s \geq (\ln 2)/24 \approx 0.0289$ per face; on 3.12.12, a twelve-support-class patch of the mid-edge system in a 3×2-cell box gives $s \geq (\ln 2)/18 \approx 0.0385$ per face; on 4.6.12, a twelve-support-class, ten-face patch (three dodecagons, three squares, four hexagons) in a 3×2-cell box (six faces per cell) gives $s \geq (\ln 2)/36 \approx 0.0193$ per face. In all three systems the 4×4-window region kernel is four-dimensional, the same translate count $(4-2)^2$ as on the honeycomb, and both completions of every patch satisfy the split-face equations exactly, rational and √3 parts separately (Fraction arithmetic). Certificates: Fig. S8b–d.

**The binary strip construction.** Values in {1,2}; the counting domain is the open strip of width W: faces (i, j), $i \in [0, L)$ open in x (boundary edges are single-face legal edges), $j \in [0, W)$ periodic in y. Eliminating the face relations $v_3 - v_0 = v_5 - v_2 = v_1 - v_4$ organises the edges into three stripe families: L column stripes (closed in y), and W row stripes and W diagonal stripes (open in x). Call a signed field admissible if every column stripe closes at net charge zero and every stripe's running walk has range at most one; an admissible field is then realised by an assignment whose values occupy, stripe

by stripe, two adjacent integers, which the all-1 base absorbs into {1,2}. A site t = (p, q) with p ≥ 2 carries the third-difference field a = $\Delta_{o0}\Delta_{o1}\Delta_{o2}\delta$_t with exactly six charges ±1,

(p,q): +1; (p,q+1): −1; (p−1,q): −1; (p−2,q+1): +1; (p−1,q+2): +1; (p−2,q+2): −1 (y mod W),

and on every stripe it touches it contributes exactly one (+1, −1) pair: a sign interval on the walk, of length one or two, raising the running sum by one (type U, plus-then-minus) or lowering it (type V).

**Lemma S8.3.4 (interval criterion).** Let S be a site set whose induced sign intervals on every stripe satisfy (a) same-direction intervals are pairwise disjoint, and (b) opposite-direction intervals are pairwise nested. Then for every subset T ⊆ S the summed field a_T is admissible.

*Proof.* On a stripe the running partial sum of a_T is the signed coverage P(x) = #{U ∋ x} − #{V ∋ x}. By (a), any position is covered by at most one interval of each direction, so P ∈ {−1, 0, +1}. If P(x) = +1 and P(y) = −1, take U ∋ x and V ∋ y; by (b) either U ⊆ V, whence x ∈ V and P(x) ≤ 0, or V ⊆ U, whence y ∈ U and P(y) ≥ 0: both impossible. The running sum therefore takes at most two adjacent values, so its range is at most one; on the closed column stripes every site contributes net charge zero, so the walk closes. ■

**Lemma S8.3.5 (three site sets).** For every width W ≥ 3 there is a site set S satisfying the interval criterion with |S| ≥ (2/25)·LW − O(W):

(i) W even: S = {(i, j): i, j odd}, the odd-odd checkerboard; density → 1/4;
(ii) W odd, W ≥ 5: S = {(i, j): i, j odd, (i + j) mod W ∈ R} with R = {0, 2, 4, …, W − 3}; density $(W-1)^2/(8W^2)$, minimal 2/25 at W = 5;
(iii) W = 3: S = {(i, 1): i ≡ 0 (mod 3)}; density 1/9.

*Proof.* (i) At even W parity is conserved along all three stripe families. On a row of odd j, the contributing sites (q = j and q = j − 2) both give V-type unit intervals with odd starts, pairwise disjoint; on a row of even j the single contributing family (q = j − 1) gives pairwise disjoint length-2 U intervals. Odd columns receive only unit U intervals, one per residue, tiling the column; even columns receive only disjoint length-2 V intervals. On the diagonals, a site with p, q odd sends its U-type interval to stripes of one parity and its V-type intervals to the other, so each stripe carries a single direction, and two sites on it differ by an even, nonzero Δp ≥ 2, so intervals of length at most two are disjoint. Condition (b) is vacuous: every stripe carries one direction only.

(ii) All pairwise offsets are even, (d, e) = (2a, 2b), with e ∈ [−(W−3), W−3] (the sites occupy the odd rows 1…W−2); the congruence e ≡ ±1 (mod W) is therefore excluded, its only in-range representatives ±1 and W ∓ 1 being respectively odd and out of range, and the same holds for d + e = r − r′ with r, r′ ∈ R, again even and confined to [−(W−3), W−3]. Three kinds of encounter are possible. *Columns:* a site (p, q) contributes a unit U interval to column p, a shifted unit U interval to column p − 2, and a length-2 V interval to column p − 1; two sites meet on a column stripe only at d ∈ {0, ±2},

where the meeting is U-against-U (same direction; the V intervals live on even columns, met by odd p never), and the supports are disjoint: at $d = 0$ the distinct odd q, q′ give non-adjacent unit cells on the cycle, at $d = \pm 2$ the shift places the cells apart unless $e \equiv \pm 1$, excluded. *Rows:* contributing sites satisfy $q - j \equiv 0, -1, -2 \pmod W$, and the interval direction is determined by this offset (V, U, V); two sites therefore contribute opposite directions only if their offsets differ by one, which forces $e \equiv \pm 1 \pmod W$, excluded, while same-direction supports are separated by the even, nonzero column offset. *Diagonals:* a site's three interval types sit at offsets $\delta = 0$ (U, support $[p-2, p)$), $\delta = +1$ and $\delta = -1$ (both V) from its diagonal index $p + q$, and two sites meet exactly when $d + e \equiv \delta' - \delta \pmod W$. Opposite direction needs $d + e \equiv \pm 1 \pmod W$, excluded above; same-direction meetings are U-U at $d + e \equiv 0$ (distinct sites have $d \neq 0$, so supports $[p-2, p)$ with even offset $\geq 2$ are disjoint) and V-V at $d + e \equiv \pm 2$ (supports separated by the even offset; at $d = 0$ the two unit V intervals occupy adjacent cells). ■

(iii) The offsets between two sites are $(3m, 0)$: all three diagonal encounters have $d + e \equiv 0 \pmod 3$, same-direction with even, nonzero d; rows are same-direction and disjoint; columns are not shared. ■

**Theorem S8.3.6 (binary honeycomb, strip bound).** For every width $W \geq 3$ and length $L \geq 4$, with S the site set of Lemma S8.3.5,

$\Omega_\text{strip}(L, W) \geq 2^{|S|} \geq 2^{(2/25)\cdot LW - O(W)}$, hence $s \geq (2/25)\cdot \ln 2 \approx 0.0554$ per face,

and along the even-width subsequence the construction gives $s \geq (\ln 2)/4 \approx 0.1731$.

*Proof.* Every subset $T \subseteq S$ has an admissible summed field (Lemma S8.3.4), and an admissible field is realised by a configuration with values in {1,2} (the two-adjacent-value window of each stripe, absorbed by the all-1 base). The map $T \mapsto a_T$ is injective: $\Delta_{o2} g = 0$ forces g to be invariant under x-translation, hence zero on x-finitely supported fields, and the two remaining differences propagate injectivity the same way; all charges of every site module lie inside the strip, so strip fields are plane fields. Distinct subsets therefore give distinct configurations, and $\Omega_\text{strip} \geq 2^{|S|}$; the uniform density 2/25 is case (ii) at $W = 5$, the O(W) term is the box-edge correction, and the even-width density is case (i). ■

*Layered reading.* The constructive bounds and the numerical determinations are two layers, stated side by side and not to be merged:

| carrier (alphabet) | constructive [T] | numerical determination |
|---|---|---|
| honeycomb ({1,2,3}) | $\geq 0.0770$ | 0.544 (fit band 0.53–0.58, S6.2) |
| 3.4.6.4 ({1,2,3}) | $\geq 0.0289$ | 0.040 (protocol 0.031–0.041, S7.2) |
| 3.12.12 ({1,2,3}) | $\geq 0.0385$ | $s_3 \approx 0.186$ (S5.3) |
| 4.6.12 ({1,2,3}) | $\geq 0.0193$ | $s_2 \geq 0.186$ at $W = 3$ [strict] (S8.2) |
| honeycomb ({1,2}) | $\geq 0.0554$ ($\geq 0.1731$ even widths) | 0.209 pinned, 0.248 free (S6.2) |
| honeycomb ({1,2,4}) | none | 0.224 (band 0.17–0.27, S6.2) |

The bounds are not tight: they sit a factor 1.4–7 below the determinations (closest on 3.4.6.4), and the numerical values remain the determinations of record. Coverage is stated exactly: the ternary alphabet is certified on all four carriers, the binary alphabet on the honeycomb; no constructive bound is claimed for {1,2,4} or for the binary split-row carriers, whose strict numerical positivity stands on the transfer-matrix rates (S8.2).

*Local versus non-local.* The two mechanisms are two completions of one object. The site module of the strip construction is the same third-difference field whose ternary incarnation is the flower flip; with three values available the field completes locally (the alphabet absorbs the compact bump), with two values the completion is forced to flatten entire stripes, so the elementary binary excitation is a whole stripe, not a patch. This is the constructive counterpart of the observation that the binary and non-arithmetic honeycomb, and the binary 3.4.6.4 and 4.6.12 systems, admit no compact two-completion certificate (their window kernel modes are all mixed-sign), and it records a dichotomy conjecture: local flips exist precisely when the alphabet contains a three-term arithmetic progression (S6.3). The conjecture is open; the bound of Theorem S8.3.6 does not depend on it.

*Verification.* The ternary certificates are checked by exact machine arithmetic: both completions of every patch satisfy all face equations (rational and $\sqrt{3}$ parts separately), take alphabet values, and vanish on the boundary; the constraint pipelines reproduce the registered kernel dimensions at anchor sizes (honeycomb $L^2+2$ at $L = 2\ldots5$; 3.4.6.4 $L^2+3$; 3.12.12 $L^2+3L+1$; 4.6.12 $4N+4$), and the square-lattice propagation system, run through the same window pipeline, returns kernel dimension zero (a zero-entropy system admits no compact mode). The strip construction is verified at four levels: dual-algorithm strip counting (edge-variable DFS against the stripe-formula count, bitwise at (L, W) = (2,2)/(3,2)/(2,3) = 72/156/358, seven further cases against a pruned DFS, five against torus anchors); the interval criterion against direct walk simulation over the full offset window (W = 5, 6, 8); the three site sets at W = 3…24 (L = 4W), per stripe and pair by pair; all $2^{14}$ subsets of the W = 3…9 site sets admissible and pairwise distinct, with end-to-end count tests $\Omega_\text{strip} \geq 2^{|S|}$ at six sizes (e.g. (6,4): 128 832 ≥ 16); and the negative control that the bare odd-width checkerboard, without the residue restriction, dies exactly at the offsets (W ± 1, 0) that case (ii) excludes. Scripts and records: Reproducibility Supplement (Tables R1, R2).

*S8.4 Open questions*

Q1 *When does "one hexagon, one module" survive?* The deficiency criterion that would turn Table S2 into a predictive rule (the honeycomb and 3.4.6.4 carry it; the kagome kills it with triangles). Q2 *When are the integer points thick?* The $\mathbb{Z}$-counting counterpart that separates the honeycomb (area law) from 3.4.6.4 (locked leak) and 4.8.8 (split rows); S6.3 adds its domain side (the count also reads the arithmetic of the sequence (coprime versus non-coprime)), and S8.1 L4′ adds its method side (transient exact sub-laws); the honeycomb binary reversal (S6.2) and 4.6.12's falsified perimeter law

(S8.2) are the newest data for this question, and the constructive certificates of S8.3 are its positive half. Q3 *The 4.6.12 asymptote*: the perimeter-law route to zero entropy is falsified at strict level, the finite-width readings are strictly positive (S8.2), and the positivity is constructive at the ternary alphabet (Theorem S8.3.3); open are the asymptote itself (three-width reference ≈ 0.09–0.10 per face) and a binary constructive bound. Q4 *Phason elasticity*: a strain theory for discrete-metric random tilings, against which the measured displacement anisotropy (Sect. S10) could be tested. (The displaced earlier Q3, the all-rational mixed-face family, was closed: all computed members are zero-entropy.) Q5 *Local versus non-local*: do strip-type site constructions certify the binary split-row carriers as Theorem S8.3.6 certifies the binary honeycomb, and is the three-term-progression dichotomy of S8.3 a theorem?

## S9. The value-domain ensemble suite: protocols and calibres

*Ensembles.* Six exact-uniform value domains on the honeycomb network: the base pair {1,2}, {1,2,3} (the binary and arithmetic affine shape classes of Sect. S6) and the arithmetic translation ladder {2,3,4}, {3,4,5}, {4,5,6}, {5,6,7}, extended for this revision to {6,7,8} and {7,8,9} (ladder extension, below). The ladder spans the same shape class as {1,2,3}, pointwise-equal in counts by the translation bijection (Lemma 2, S7.3; tile counts asserted at run time: 45 for every arithmetic domain, 10 for {1,2}). The non-arithmetic domain {1,2,4} is deferred at this calibre: its exact uniform ensemble is identified, the reflection identity $\langle l \rangle = \mathrm{mean}(A) = 2.3333$ verified by exhaustive enumeration at 3×3, 4×3, 4×4 and 5×4 ($\Omega$ = 19 707, 6 561, 534 789, 59 049), and the sampler is unbiased on it (small-system tests within 1$\sigma$; four independent production-size runs scatter by ±2% in $\langle l \rangle$ with no systematic offset), but that scatter is an order of magnitude above the arithmetic domains' and the sampler's mixing at the production torus is therefore not yet at the suite's calibre for gapped alphabets; its counting-layer classification (S6) is unaffected. For each domain, 20 closure-respecting configurations at N = 169 holes (the 13×13 honeycomb torus, cut open to a simply connected patch for diffraction; the cut envelope is fixed across ensembles) are generated by the two-stage exact-uniform sampler: a single parallel-tempering chain per domain (eq = 1.5M steps, snapshots thinned by 40k steps; seed family 20261100 + k, rejection-extended; per-configuration seeds archived). The twenty configurations of one domain are therefore chain snapshots, and the intervals quoted throughout are configuration-bootstrap intervals at that calibre. Validity is exact: every delivered configuration satisfies the closure conditions by construction (verified from closure_rows at run time), and the geometry is extracted by the template tiling of the cut patch (ideal-patch gate: interior vertices perfectly honeycomb at 120°).

*The ladder rungs are one sample.* The four ladder offsets are not independent samplings. The sampler's solution structure (the difference-coordinate skeleton and the valid-shift profiles) depends only on the constraint cone and on the span of the alphabet, so a single seed family delivers the same configurations for every offset, relabelled by the translation. The delivered ensembles verify this

exactly: translating the {1,2,3} ensemble by +3 reproduces the {4,5,6} ensemble to machine precision (embeddings agree to $\leq 3\times10^{-11}$), every reported statistic coincides, and the per-window displacement-spectrum ratios of the five arithmetic domains agree to $\leq 0.5\%$, the residual being the second-order reweighting of the per-configuration amplitude factors $\alpha = \ell/(\ell + c)$ (S10.4, identity 1). The ladder comparison is therefore noise-free by construction. The binary {1,2} class and the deferred {1,2,4} domain are separate samples.

*Ladder extension (this revision).* The two smoothest rungs, {6,7,8} and {7,8,9} (relative dispersion $\sigma/\mu = 0.117$ and 0.102), are exact translation relabellings of the {1,2,3} sample (x + 5 and x + 6); no new sampling is involved (the relabelled configurations are archived as `raw/torus13_a678.npz` and `raw/torus13_a789.npz`). Read through the same pipeline they give f_coh = 0.489 and 0.577, f_inc = 0.093 and 0.083, halo/core = 0.71 and 0.60, r = 0.43 and 0.47, and $S\perp/S\parallel = 0.129 \pm 0.040$ for both, identical to the five production rungs as the ladder identity of S10.4 requires. The extension verifies the Debye–Waller law of S10.4 at its smooth end (measured/predicted 0.5% and 0.2%) and brings the ladder to the smallest dispersions this shape class admits at this calibre.

*Estimators.* Structure factors are exact sums over the unit-cell contents of each configuration with a convex-hull mask, at mean-bond-length normalisation (q_max = 10.5, $601^2$ grid). The coherent fraction $f_coh = |\langle F\rangle|^2/H_{ref}$ evaluates the ensemble-averaged complex amplitude at the six second-shell spot positions (the strong {11}-type $\sqrt{3}$ shell, $|G| = 4\pi\sqrt{3}/3$ in bond-length units) against $H_{ref}$, the ideal (uniform-tile) patch evaluated by the same raw-sum estimator as the ensemble amplitudes (all vertices, per spot); the displayed $|F|^2$ images use the incircle-masked structure factor instead, whose ideal peak (H_id = 97 017.6 for the V = 384-site, 169-face disc) differs from $H_{ref}$ by the per-spot mask factor (1.51, 1.49, 1.49, 1.51, 1.49, 1.49); f_coh normalises by $H_{ref}$, not by H_id, so a uniform ensemble reads f_coh = 1 by construction. Spot positions are set by the ideal-patch reference itself (shell-annulus argmax of the ideal $|F|^2$, rotated by the six-fold symmetry, snapped to the local maximum of the grid); for the cut-disc geometry the second-shell ($\sqrt{3}$) spots sit at azimuths $0° + 60°k$, so the estimator adapts to the patch geometry rather than fixing an azimuth convention. Halo metrics integrate the ensemble-averaged $|F|^2$ in the annulus $0.22 \leq |q - G| \leq 1.10$ around each spot against the core ($|q - G| \leq 0.22$); the ratio is a raw-sum ratio of the two windows, so its flat-field (uncorrelated-intensity) value is the pixel-count ratio 24.8, and the sub-unity disordered-end readings of Table S7 are relative to that floor; the halo anisotropy r is the second-moment aspect ratio of the spot-centred cloud. The displacement field $u = r - sR$ (per-configuration optimal scale s against the ideal reference, matched site-by-site through the graph topology; site identities are alphabet-independent) is decomposed into longitudinal and transverse spectra in the halo window k = 0.22–1.10, binned at 0.08. Configuration-bootstrap intervals: B = 200 resamples of the 20 configurations.

*Calibre.* (i) **Sampler.** The production ensembles are sampled by an external two-stage exact-uniform sampler (sampler2: difference-coordinate structure theorem with sparse-combination proposals and

parallel tempering; vendored in the scripts folder with its provenance and gates). Local gates reproduced before deployment: exact count count_solutions(4,3,{1,2}) = 256; the reflection identity ⟨l⟩ = mean(A): 1.5 (binary), 2.0 (ternary), 2.3333 (non-arithmetic {1,2,4}, exact by exhaustive enumeration at 3×3, 4×3, 4×4 and 5×4); per-configuration closure verified from closure_rows at run time; swap rates logged per run. (ii) Chain-snapshot calibre: the twenty configurations of a domain are snapshots of one chain (thinned by 40k steps), so configuration-bootstrap intervals measure the spread over snapshots, not over independent chains. (iii) The incoherent share at the spot positions, f_inc = $\langle|F|^2\rangle/H_{ref}$ − f_coh, accompanies f_coh: at the base domains and the lower ladder rungs the spot intensity is fluctuation-dominated (f_inc ≫ f_coh), and coherence takes over along the ladder; at {5,6,7}, f_inc = 0.10 against f_coh = 0.38 at this size (Table S7). (iv) Envelope window: the open-patch diffraction convolves the true $|F(q)|^2$ with the fixed cut envelope; the window is common to all ensembles (same cut, same grid), the incircle mask softens it, and f_coh normalises by the ideal patch of the same envelope in the same estimator; the window divides out to first order. (v) Six-spot inhomogeneity: under the anisotropic cut the six second-shell spots split into inequivalent classes; within a domain the per-spot f_coh spreads by up to a factor of five (deeper disorder, wider spread); all quoted values are six-spot means, per-spot tables and bootstraps archived. (vi) **Binary-domain ensemble systematic.** Independent production runs of the {1,2} domain settle into distinct long-lived roughness sectors at matched calibre: the coherent fraction reads $1.8\times10^{-4}$ (one run) against $1.0\times10^{-2}$ (a second run, on the larger 21×21 ensemble), and the real-space displacement roughness reads 0.29 against 0.13 across the same two runs ($w^2$ at R = 6 on the 13×13 ensemble against the 21×21 one; the single-ensemble ladder is the main-text Fig. 4d). The arithmetic ladder domains show no such spread (≤ 4% between runs and continuity across the ensemble-size switch). The binary domain's halo readings are therefore quoted with this ensemble-level systematic attached, and the spread itself (consistent with slow mixing of the thinnest-entropy class) is recorded as an open question; the ladder results of §2.3 are unaffected. (vii) **Mirror-gate calibre (tension family).** The tension ensembles of this revision are sampled from four independent chains of five configurations per tension value (parallel tempering, eq = 1.5M steps, snapshots thinned by 40k, a denser 13-rung replica ladder than the production suite's seven; the hot-replica swap rate rises to 0.70-0.71 from 0.48-0.53 and the chain length is unchanged). The exact mirror identity of S10.4 (identity 2) is the sampler gate, and the delivered ensembles satisfy it: the per-face length sums obey m(h) + m(-h) = 12 to +0.027 +- 0.061 (|h| = 0.15) and +0.012 +- 0.026 (|h| = 0.30), each +0.44 sigma, and the value marginals $P_{+h}$(v) against the mirrored $P_{-h}$(4-v) agree within 1.5 sigma at every value and both tensions (largest deviation 0.0145 +- 0.0114 at |h| = 0.15, v = 2). The exact entropy-symmetry consequence, Delta_s(h) = Delta_s(-h), is reproduced to 1%: -0.082 / -0.083 $k_B$ per face at |h| = 0.15 and -0.490 / -0.488 at |h| = 0.30. The same two-way reading bounds the diffraction-block calibre: the twin readings of each corner agree to 4% in f_coh (0.555/0.536 at h = −0.30; 0.185/0.199 at h = +0.30), while at |h| = 0.15 they scatter by factors of 3.5-4.9 in the same direction (both below the uniform value at h = +0.15,

both above at h = −0.15), so the intermediate-tension readings are quoted as the mean of the two samplings with their difference as the uncertainty, and the corner readings as their agreement. [The earlier production runs of this block, sampled as snapshots of a single chain at the seven-rung ladder, carried an ensemble-level mirror violation (m-sum 12.050 / 11.869) and are superseded; the corner-composition scatter they showed (defect-line count histograms {13, 5, 1, 1} against {5, 8, 6, 1}) is the same finite-sample effect.]

*h-Ensembles (metric tension).* The tension family $\pi_h \propto \exp(-h \cdot \Sigma_e x_e)$ is sampled by the same parallel-tempering machinery with $h \neq 0$ (Boltzmann weights in the second stage; the shift-picker draws constants with weight $\exp(-h \cdot \text{mult} \cdot a)$, mult the line multiplicity). Protocol for the arithmetic domain {1,2,3}: $h \in \{-0.30, -0.15, +0.15, +0.30\}$ plus the h = 0 uniform ensemble, 20 configurations each at N = 169 from four independent chains per tension (calibre above). Gates: per-configuration closure (run-time); the exact mirror identity of S10.4 and its entropy-symmetry consequence (calibre above); swap and acceptance rates logged per run. From the response $m(h) = \langle \Sigma l \rangle_{h/F}$: the entropy cost $\Delta s(h) = \int_0^h h' \cdot m'(h') dh'$ (thermodynamic integration over the measured points, piecewise trapezoid including every intermediate tension, anchored at h = 0; the h = 0 reference needs no absolute ln Z; by parts this equals $h \cdot m(h) - \int_0^h m \, dh'$). Measured response: m(h) = 8.912 / 7.112 / 6.008 / 4.915 / 3.100 per face at h = −0.30 / −0.15 / 0 / +0.15 / +0.30, monotone, and Δs = −0.488 / −0.083 / 0 / −0.082 / −0.490 $k_B$ per face: the entropy cost is symmetric in the tension sign to 1% or better, as the mirror identity requires (Table S7, tension block). Phason observables (f_coh, halo/core, r) are evaluated per h exactly as for the uniform suite (Fig. 4e–g; the coherence is non-monotone in h: the biased ensemble passes through the ragged binary-mixture regime on the compression side before the near-uniform corner is reached; see main text §2.3). The tension ensembles are exact mirror pairs in the state space: $l \to 4 - l$ is a bijection of the {1,2,3} constraint set and exchanges $\exp(-h \cdot \Sigma l)$ with $\exp(+h \cdot \Sigma l)$, so the −h ensemble is the mirror image of the +h ensemble; the diffraction readout is not mirror-covariant (S10.4, identity 2), which is the origin of the coherence asymmetry at equal |h|.

## S10. Diffraction and anisotropy: results and additional data

### *S10.1 Pair-density decay*

On the delivered suite the ensemble-averaged g(r) oscillates with an RMS envelope (sliding window, 20 configurations per domain) that falls monotonically from 0.44–0.81 in the window $r \in [2,3]$ to a far-window plateau of 0.28–0.57 at $r \in [6,7.5]$ (57–77% of the mid-range value), so the translational correlation range of the metric field is a few bond lengths, with no growth over the accessible window. Across all six domains the envelope is fitted better by a power law (exponents p = 0.49–0.88, $R^2$ = 0.82–0.93) than by an exponential, consistent with the algebraic-decay expectation for two-dimensional random tilings; the incircle clipping (equivalent disc radius $R_{in} \approx 11$ bond lengths) limits the far-r range, so the two forms are not separated with finality, and the algebraic assignment is

the ensemble-supported reading. The plateau amplitude orders with the domain's relative dispersion, b12 ($\sigma/\mu = 0.333$) 0.57 against a345 (0.204) 0.28: the rougher the value domain, the larger the residual oscillation (Fig. S9a). Data: `data/uniform/pdf_decay_suite.csv`, curves `pdf_curves_suite.npz`.

*S10.2 Edge-colour structure factor $S_L(q)$*

The structure factor of the scalar edge-length field is a metric-ensemble-exclusive observable: a positional-disorder control has no discrete library to scatter from, and the permutation null (length labels shuffled within the same geometry) is flat by construction (measured flat to ±8% across $q \leq 9$). Against it the delivered suite shows a three-regime structure (Fig. S9b). The low-q wing ($q \lesssim 2$) sits at the null level for the arithmetic domains (ratio 0.94–1.09) while the binary domain roughly doubles it (2.0 in $q \in [0.1, 0.5]$), the phase-separation signature of the thinnest-entropy class. A mid-q band carries enhanced metric-field weight over the null (up to 1.6 × at $q \in [4, 6)$ for the arithmetic domains). Across the second-shell region ($q \in [6, 8]$, around the strong {11}-type Bragg shell at $|q| = 7.26$; the honeycomb Bragg shells lie at $|q| = 4.19$, 7.26 and 8.38 in units of $1/\ell$, the weak {10}-type first shell falling inside the mid-q band above) the diffuse floor is suppressed to 0.48–0.63 of the null once the bright spot cores are masked by the data-driven, shell-agnostic spot detector (local maxima above 20% of the map maximum, mask radius 0.6): the metric field's spectral weight at those wavevectors is concentrated in the coherent peaks rather than spread over the floor. In principle $S_L(q)$ is the discriminator against paracrystalline disorder, which has a continuous lattice-vector distribution (hence no discrete motif library or matching rules) and fixes only an average local orientation: its angular fluctuations make the embedded $\psi_6$ fall below unity rather than being identically one (Table S1). The paracrystal ensemble itself is not simulated in this paper, so the separation is structural rather than measured. Data: `data/uniform/slq_suite.csv`, profiles `slq_profiles_suite.npz`.

*S10.3 Halo anisotropy and the displacement spectrum*

*Halo shape.* Each second-shell spot of the metric ensemble carries an anisotropic diffuse cloud. On the S9 suite the halo aspect $r = \sigma\perp/\sigma\parallel$ (ensemble-averaged $|F|^2$, second-moment protocol) reads 0.36 [configuration-bootstrap 0.22–0.63] for {1,2}, 0.29 [0.22–0.73] for {1,2,3}, and 0.30 / 0.34 / 0.37 / 0.40 for the ladder offsets {2,3,4} / {3,4,5} / {4,5,6} / {5,6,7}, below unity for every domain of the suite, the radial halo elongation that $S\perp/S\parallel < 1$ requires (Table S7; main text Fig. 4c); the fixed-shape phason-elasticity literature[S10–12] supplies the comparison class for this kind of diffuse-cloud readout. [An earlier, differently constructed {4,5,6} ensemble (vertex-constrained, growth-solver sampling) read r = 0.42–0.52 (halo-window average, 20 configurations); the ideal-lattice control reads 0.99 ± 0.00.] Two-layer reading (Fig. S10): the principal axes of *single-configuration* halos scatter widely where the halo is speckle-dominated ($\Delta\theta$ s.d. 35.0° ({1,2}) and 38.6° ({1,2,3}) against the uniform-theory 52°) and tighten toward the radial direction along the arithmetic ladder (14.4° / 11.6° / 10.9° / 10.7° for {2,3,4} / {3,4,5} / {4,5,6} / {5,6,7}); the *ensemble-averaged* axes lock onto each spot's own

reciprocal vector (mean $|\Delta\theta| = 2.2–2.9°$ across the arithmetic domains, 12.7° for the binary; the binary's single-configuration scatter is the widest of the suite, 35°, reflecting its speckle-dominated halo). The diffuse orientation belongs to the average structure; the single-configuration residual is the speckle of the disorder, and its shrinkage with coherence follows the same ordering axis as the halo weight and shape (main text Fig. 4b,c). Shell indexing: the halo readouts are taken at the strong {11}-type second shell ($|G| = 4\pi\sqrt{3}/3$); the crystallographic first shell ({10}-type, $|G| = 4\pi/3$) carries a four-times smaller per-cell coherent weight and a Debye–Waller exponent three times smaller, so it loses coherence three times more slowly under the same disorder. The first shell appears in Fig. 4a as the inner hexagon of features rotated by 30° from the marked positions, degrading into radial streaks at the disordered end; in a real material the innermost {10} spots would retain coherence longest, a complementary experimental fingerprint.

*Displacement field and the anisotropy ratio.* The displacement field of the S9 protocol ($u = r - sR$ against the ideal reference, graph-topology matching) is decomposed against each wavevector as $u_\parallel = (u\cdot\hat{k})\hat{k}$ and $u_\perp = u - u_\parallel$, with spectra $S_\parallel(k) = \langle|u_\parallel(k)|^2\rangle$, $S_\perp(k) = \langle|u_\perp(k)|^2\rangle$ averaged over the quenched metric disorder, in the phason convention of Socolar et al.[S13] The ratio is a direct, dimensionless measure of the fluctuation anisotropy of the metric field; no interpretation in terms of elastic constants is offered here (a strain theory of discrete-metric tilings that would supply one does not yet exist; S8.4 Q4). On the S9 suite the halo-window ratio reads $0.169 \pm 0.056$ for the binary {1,2} and $0.136 \pm 0.049$ on each of the five arithmetic domains, 0.14–0.17 across the six exact-uniform domains (main text Fig. 4c; per-bin data in the archived spectra table). The statistics run over the ten resolved bins of the halo window, $k = 0.34–1.06$: the innermost bin ($k = 0.22–0.30$) lies entirely below the 13×13 supercell's fundamental Bloch shell, $|k|_{min} = 4\pi/(\sqrt{3}\cdot d\cdot 13) = 0.322$ ($d = \sqrt{3}$, the centre spacing in units of the mean bond length), so it carries no resolved mode of the simulated box; it reads high in every domain (0.50 for the five arithmetic domains, 0.61 for the binary) and its inclusion would govern the mean through a single bin that carries no resolved mode of the box; it is excluded, identically for all domains. Within the retained window the ratio drifts downwards with k by a factor of 2.7 (per-domain slope $-0.13 \pm 0.05$ per unit k, $2.4\sigma$), so the quoted mean is a window average and the drift itself is measured content. The arithmetic domains' ratios coincide to $\leq 0.5\%$ by construction (S9 ladder identity); more strongly, the ratio is invariant under any global rescaling of the displacement field (a ratio of two components at one wavevector cannot see a scale factor), and the ladder translation is exactly such a rescaling (S10.4, identity 1). The independent diffraction-space route, the halo shape anisotropy r (halo-shape paragraph above), lands on the same side of unity for every domain of the suite ($r = 0.29–0.40$, six-spot means, bootstrap 95% intervals excluding unity throughout; Table S7), as the transverse-depleted spectrum requires for a radially elongated cloud.

*S10.4 Kinematic identities of the ladder and the tension mirror*

Two exact identities of the diffraction pipeline organise the tension and ladder readings of the main text; both were verified per configuration to machine precision ($\leq 1\times10^{-11}$) on the delivered ensembles.

*Identity 1: the ladder is an amplitude flow of the displacement field.* Let x be a configuration and x + c its translate along the ladder (c constant on every edge). The embedding is affine in the assignment ($r(x + c) = r(x) + c\cdot R_id$ with R_id the ideal all-ones embedding), and the mean bond length shifts exactly, $bl(x + c) = bl(x) + c$. Under the mean-bond normalisation of the pipeline (positions divided by bl, centroid aligned, optimal scale against the ideal reference), the displacement field therefore scales by a single per-configuration factor:

$$u(x + c) = \alpha\cdot u(x),\quad \alpha = \ell/(\ell + c),\quad \ell = bl(x).$$

Consequences: (i) any ratio of two quadratic components of the field at the same wavevector is *exactly* invariant along the ladder; this is why S⊥/S∥ is value-domain independent within the arithmetic shape class (the measured per-window ratios of the five arithmetic domains agree to ≤ 0.5%, and the extended rungs {6,7,8} and {7,8,9} reproduce the same ratio, 0.129 ± 0.040); (ii) amplitude-sensitive reads (f_coh, I_halo/I_core, the halo aspect r) drift along the ladder by construction, tracking the single scalar α; (iii) the coherent fraction follows the Debye–Waller law of the flow. With $|G|^2 = 52.638$ in bond-length units and the relative dispersion σ/μ of the value domain (proportional to α to 0.13% across the ladder), the measured f_coh equals $\exp(-|G|^2(\sigma/\mu)^2)$ at ratios 1.23 / 1.05 / 1.02 / 1.01 of prediction on the {2,3,4}…{5,6,7} rungs and 1.005 / 0.998 on the extended rungs {6,7,8} / {7,8,9}; the most disordered rung reads 5.7× above the law. At that end the excess over the law shrinks with the patch size, from 7.5× at N = 127 to 2.0× at N = 331 (S10.5), so the fixed-size span of the ladder in the main text is a lower bound on the asymptotic coherent-fraction contrast between its smooth and disordered ends.

*Identity 2: compression and stretch are exact mirror ensembles.* The map $\sigma: l \to 4 - l$ is a bijection of the {1,2,3} constraint set: the per-face closure rows are homogeneous (their coefficients sum to zero, and the constant assignment $l \equiv 2$ satisfies them), so σ maps solutions to solutions; the sampler's difference-coordinate conditions are odd under $\delta \to -\delta$ and are likewise preserved; and the alphabet maps to itself. The tension weight transforms as $\Sigma l(\sigma(x)) = 12F - \Sigma l(x)$ per torus, so $\exp(-h\cdot\Sigma l(\sigma(x)))$ $\propto \exp(+h\cdot\Sigma l(x))$: the −h ensemble is the mirror image of the +h ensemble, exactly. The diffraction readout, however, is **not mirror-covariant**, because the mirror is not a similarity: in the mean-bond normalisation the displacement field transforms as

$$u(\sigma(x)) = -\kappa\cdot u(x),\quad \kappa = \ell/(4 - \ell),$$

with $\kappa \approx 2.95$ at $|h| = 0.30$ and $\kappa \approx 1.37$ at $|h| = 0.15$. The same absolute fluctuation is therefore read against a smaller mean spacing on the compressed side (the relative dispersion is inflated by κ), and since f_coh and the halo weight are controlled by the relative dispersion (S9; main text Fig. 4b),

compression must read rougher than stretch at equal |h|. Re-reading the delivered runs through the mirror quantifies it: the h = −0.30 ensemble, mirrored, reads f_coh = 0.536 and I_halo/I_core = 0.63 (against 0.555 and 0.61 direct), while the h = +0.30 ensemble, mirrored, reads 0.199 and 1.43 (against 0.185 and 1.41 direct). The two independent readings of each corner therefore agree to 4% in f_coh, and the measured corner asymmetry of Fig. 4g is the mirror geometry itself; the m(h) and Δs(h) responses, being functions of the absolute lengths, stay symmetric.

*S10.5 Patch-size scaling of the diffraction readings*

The single-size calibre of the suite is lifted by a patch-size series on the archived 21×21 ensembles (`raw/tor21_a123.npz`, `tor21_a456.npz`, `tor21_b12.npz`), each read through hexagonal discs of radius R = 6…10 (N = 3R(R+1)+1 = 127, 169, 217, 271, 331 faces) with the same estimator and the same cut type (`vd_ladder_ext.py size`).

All readouts are size-dependent to some degree, and the dependence organises by the disorder strength:

| **domain (σ/μ)** | **f_coh, N = 127 → 331** | **halo/core, N = 127 → 331** | **r, N = 127 → 331** |
|---|---|---|---|
| {1,2,3} (0.408) | $1.18\times10^{-3} \rightarrow 3.2\times10^{-4}$ (×0.27; flat beyond N = 271) | 14.7 → 15.1 (≤ 5%) | 0.29 → 0.34 |
| {4,5,6} (0.163) | 0.281 → 0.195 (×0.69) | 1.31 → 0.92 | 0.37 → 0.30 |
| {1,2} (0.333) | $1.1\times10^{-2} \rightarrow 5.9\times10^{-3}$ (×0.55) | 3.7 → 2.4 | 0.33 → 0.28 |

The coherent fraction falls with patch size in every domain, steeply at the disordered end and gently at the smooth end. The Debye–Waller law of S10.4 (identity 1, consequence iii) is approached from above as the patch grows: the {1,2,3} excess over the law falls from 7.5× at N = 127 to 2.0× at N = 331, so the fixed-size (N = 169) span of the main text is a lower bound on the asymptotic coherent-fraction contrast. The halo-to-core ratio is size-stable at the disordered end (≤ 5% drift) but falls by 30% at the coherent end, where the fixed core window captures a size-dependent share of the narrowing coherent peak; the halo aspect r drifts weakly toward isotropy with size (0.29–0.37 → 0.28–0.34). All main-text comparisons are at the fixed suite size N = 169.

## S11. Computational register

The constructive lower-bound certificates of S8.3 (four ternary flip patches and the binary strip construction) are verified by standalone exact checkers, registered in the same Supplement. The counting machinery is organised in three anchored tiers: exact enumeration (DFS with integer kernel parametrisation, reach F ≲ 20; the verification-anchor tier); transfer matrices, traces and exact elimination (the exact workhorse: keyed dynamic programs with states merged by key before vectorisation, strip operators counted by iteration or by the exact trace $\mathrm{Tr}(T^{Lx})$, cylinder spectral radii on explicit sparse operators with ARPACK under a four-gate discipline — machine-bitwise anchors, head-side purity, tail connectivity, known-answer calibration — and constraint-space elimination with

exact rational back-substitution for face-constraint systems); and calibrated stochastic estimators (Knuth tree sampling, Hutchinson trace), deployed only where exact counts cap out and only above an anchor ladder. Exact arithmetic in constructed number fields, run-time symmetry asserts, registered out-of-sample predictions for expensive extrapolations, and fail-closed allocation budgets apply throughout; power iteration is inadmissible for these operators' degenerate leading spectra. **The engine table, the file map, and the full verification protocol are tabulated in the separate Reproducibility Supplement (Tables R1 and R2 there).**

*Mixing versus metric: the entropy split at fixed composition (Fig. S11).* Because the alphabet plays three roles at once (S1.2), the entropy counted in this paper could be mistaken for an alloy mixing entropy, with the geometric content reduced to a correction. The relation is the opposite way round, and it can be stated exactly.

*Exact identity (uniform measure).* Condition on the composition vector N (the counts of each length value) and write $\Omega_N$ for the number of valid configurations with composition N. For any ensemble of valid configurations,

$$\ln \Omega = H_{comp} + \langle \ln \Omega_N \rangle, \text{ with } H_{comp} = -\Sigma_N P(N) \ln P(N),$$

where the average is over the induced composition distribution. The second term, the conditional count of closure-respecting configurations at fixed composition, is the quantity this paper evaluates: its growth rate per face is $s_\infty$, and the exact enumeration series reproduces the transfer-matrix values along the size sequence (Fig. S11a). At fixed composition there is no assignment freedom left for the lengths; what remains is a purely geometric count of how many closed tilings realize that composition. The first term is the composition-distribution entropy, the channel through which any mixing-like contribution would enter; it is bounded (it grows at most polynomially in the system size, so its entropy density vanishes) and it is not the dominant term in the measured range.

*Comparison with the alloy bound.* The natural reference is the ideal mixing entropy of a random alloy at the same composition, E edges carrying k equiprobable values: $S_{mix}/F = (E/F)\cdot \ln k = 3 \ln k$ per face on the honeycomb (2.08 for k = 2, 3.30 for k = 3). The measured entropy lies *below* this bound at every size and the gap grows with the counting domain (Fig. S11a): at F = 15 the ternary honeycomb measures 0.81 against the bound 3.30, and the closure cost rises from 2.14 at F = 6 to 2.49 at F = 15. The decisive test is the matched-alphabet comparison (Fig. S11b): {1,2,3} and {1,2,4} share k and the same bound to the digit, yet measure 0.81 against 0.66 at F = 15, a 23% difference that is arithmetic (the resonant midpoint of {1,2,3} against the non-resonant {1,2,4}) and cannot arise from a mixing channel that depends on k alone. The geometry is not a small correction on top of mixing; it is what sets the entropy, and the split above is what separates the two cleanly. Data: `si_entropy_split.csv` (exact enumeration; the identity is asserted digit-for-digit at every point); figure `S11_entropy_split.*`.

For a material context: a framework's stoichiometry is fixed at synthesis, so the composition is not a fluctuating variable, and the mixing-like channel is not the one an experiment controls. The quantity counted here is what the geometry allows at that fixed composition.

## Supplementary references


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## Figures

*Figs. S9 and S10 are drawn from the delivered value-domain ensemble suite (Sect. S9; regenerated 2026-10-13): S10.1/S10.2 readings and the halo-orientation statistics are computed on the six exact-uniform domains (20 configurations each), not on the earlier vertex-constrained ensembles. The real-space displacement-roughness ladder of the suite lives in the main text (Fig. 4d: $w_{rms}$ vs patch radius on a single 21×21 ensemble across the full range); the binary domain's ensemble-level roughness*

*spread remains a data-level record of S9 calibre (vii) ($w^2$ at R = 6: 0.29 against 0.13 between the two production runs).*

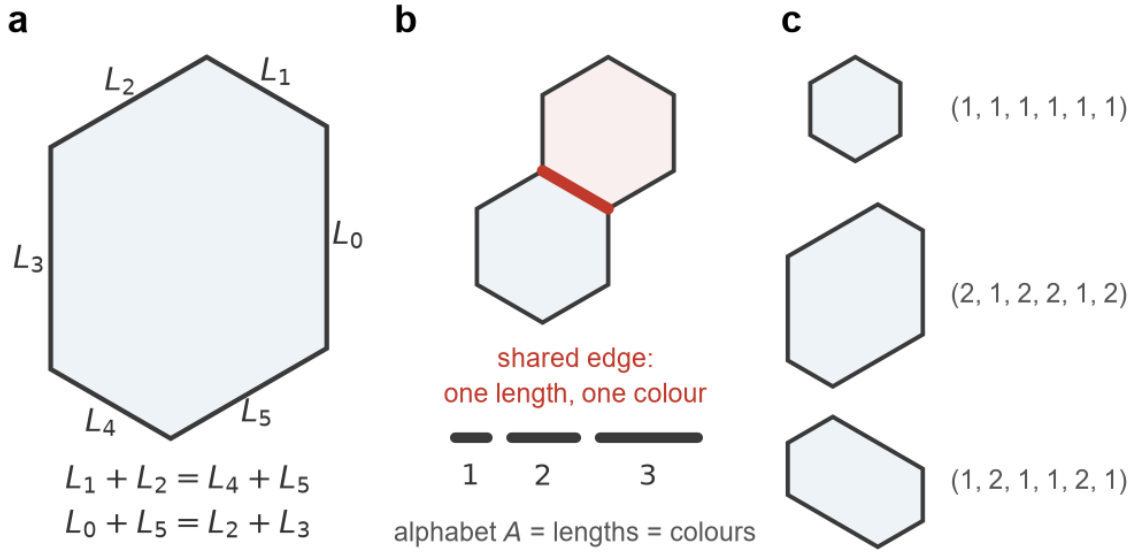



**Fig. S1 | The model in one figure. a**, An equiangular hexagon with directed edge lengths $L_0 \ldots L_5$; closure reduces to the two opposite-side sum rules. **b**, Two tiles share an edge: the shared edge has one length and one colour, the Wang matching condition; the alphabet strip {1,2,3} shows lengths drawn to scale. **c**, Three closed hexagons at identical (120°) angles with different length profiles (1,1,1,1,1,1), (2,1,2,2,1,2), (1,2,1,1,2,1): a four-parameter shape family per tile at fixed combinatorics.

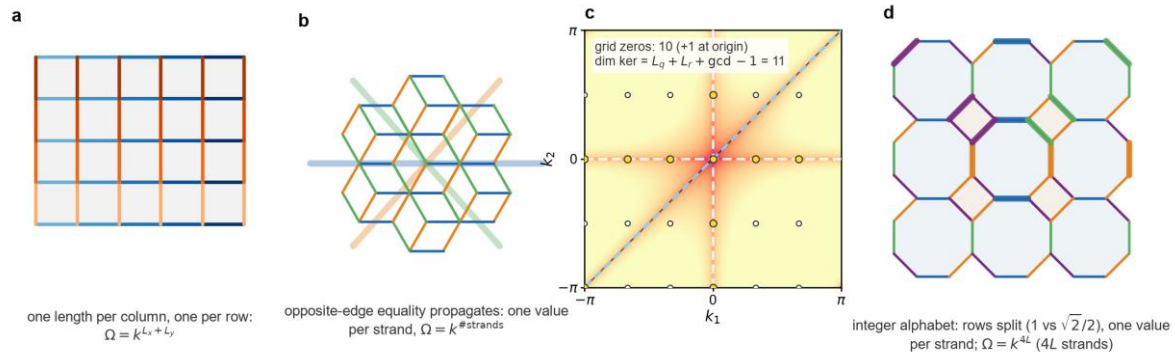



**Fig. S2 | The three zero-entropy mechanisms, one panel each. a**, Propagation (square lattice): every horizontal edge in a column carries one length (blue shades, one per column) and every vertical edge in a row one length (orange shades, one per row); $\Omega = k^{Lx+Ly}$, perimeter not area. **b**, Strip theorem (rhombic patch, drawn periodic): opposite-edge equality propagates strand-by-strand (three strand families coloured; thick lines: exemplar strands); $\Omega = k^{\#strands}$. **c**, Spectral mechanism (kagome): $|\det \hat{M}(k)|$ over the Brillouin zone (log colour) with its exact zero lines $k_1=0$, $k_2=0$, $k_1=k_2$ (dashed); the Lq=6, Lr=4 momentum grid meets the zeros in 10 points (+1 double at the origin); dim ker = Lq+Lr+gcd−1 = 11. **d**, Split rows (4.8.8): edges coloured by their four direction families (thick: exemplar equality chains, i.e. straight runs for the horizontal/vertical families, staircase chains through the squares for the diagonals); with an integer alphabet the octagon equations split (coefficients 1 vs √2/2) and every constraint becomes an equality: one value per strand, $\Omega = k^{4L}$.

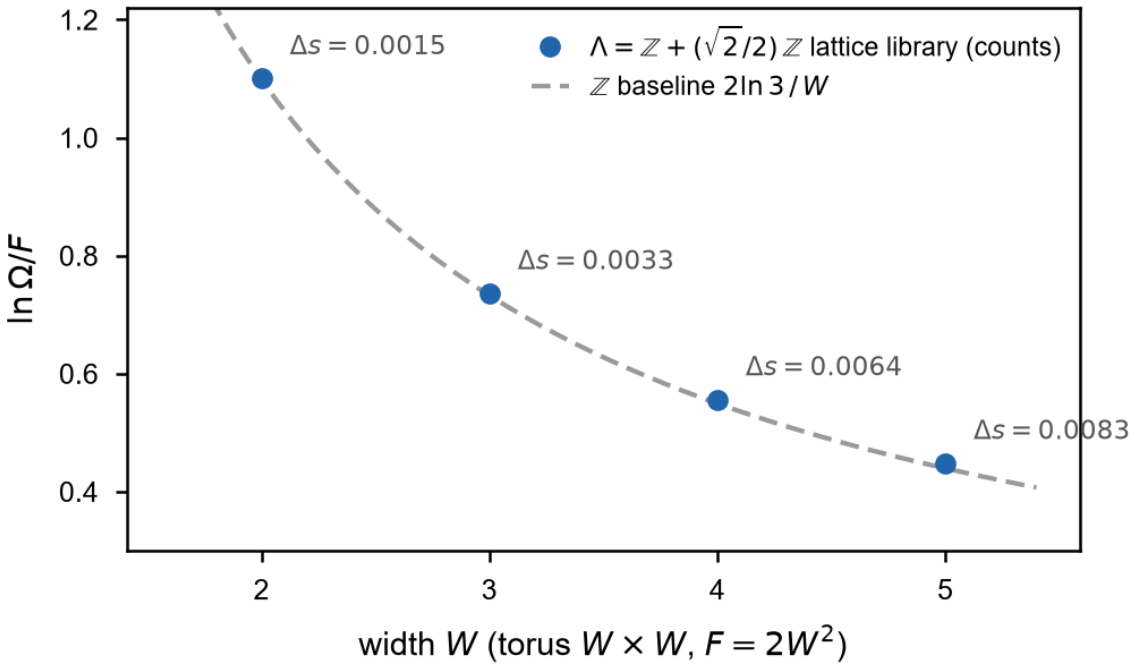



**Fig. S3 | The ℚ(√2) lattice library counts hug the ℤ baseline.** ln Ω/F for the Λ library (points, exact at W = 2–4, stochastic trace at W = 5) against the ℤ-library baseline 2ln3/W (dashed). The excess Δs = 0.0015/0.0033/0.0064/0.0083 stays at the $10^{-3}$–$10^{-2}$ level while ln Ω/F falls from 1.10 to 0.45: the counting stays sub-extensive.

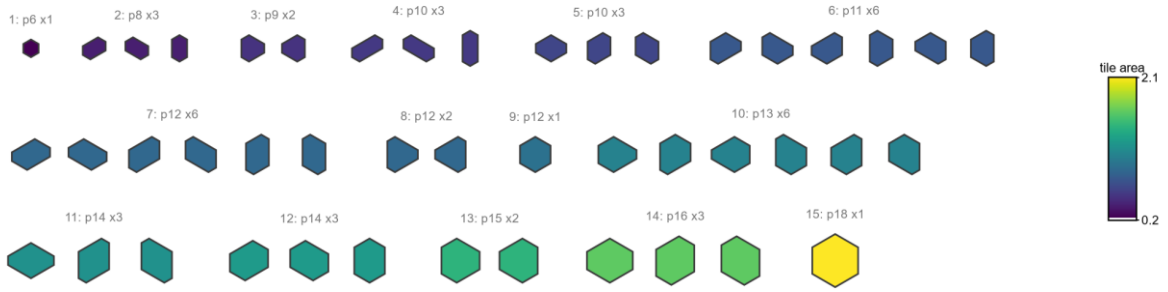



**Fig. S4 | Tile-library enumeration on {1,2,3}.** All 45 closed hexagons of the alphabet, grouped into their 15 dihedral orbits (orbit index, perimeter p, member count m; faces coloured by tile area). This enumeration is the exact basis of the tile libraries drawn in Fig. 3b; the {1,2,4} library (33 hexagons, 13 orbits) is drawn in full in Fig. 3b.

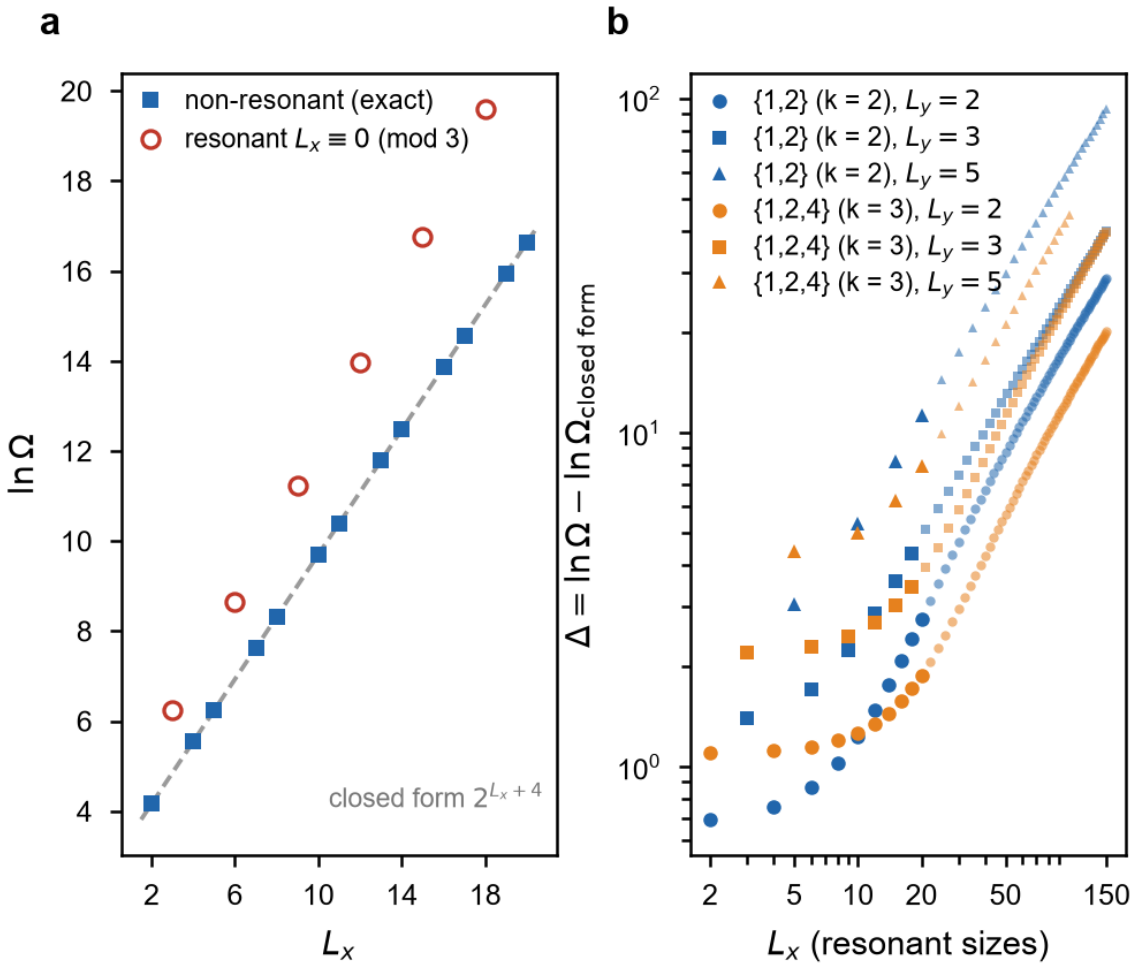



**Fig. S5 | Resonant winding sectors: where the metric entropy lives. a**, ln Ω vs $L_x$ for the binary class at $L_y = 3$: non-resonant sizes sit on the closed form $2^{L_x+4}$ (grey dashes); resonant sizes ($L_x \equiv 0$ mod 3, open red) peel above it with growing excess. **b**, Resonant excess $\Delta = \ln \Omega - \ln \Omega_{closed\text{-}form}$ vs $L_x$ for both sub-law classes and widths $L_y = 2, 3, 5$ (log–log; colour = class, marker = width; large symbols: exact counts to $L_x = 20$; small light symbols: exact long-chain counts to $L_x = 150$): Δ grows with $L_x$ at every width, through the crossover and into the asymptotic regime: the winding sectors scale with system size, and their width asymptotes $\lambda(L_y)/L_y$ (Fig. 3e) are the entropy itself, not an add-on.

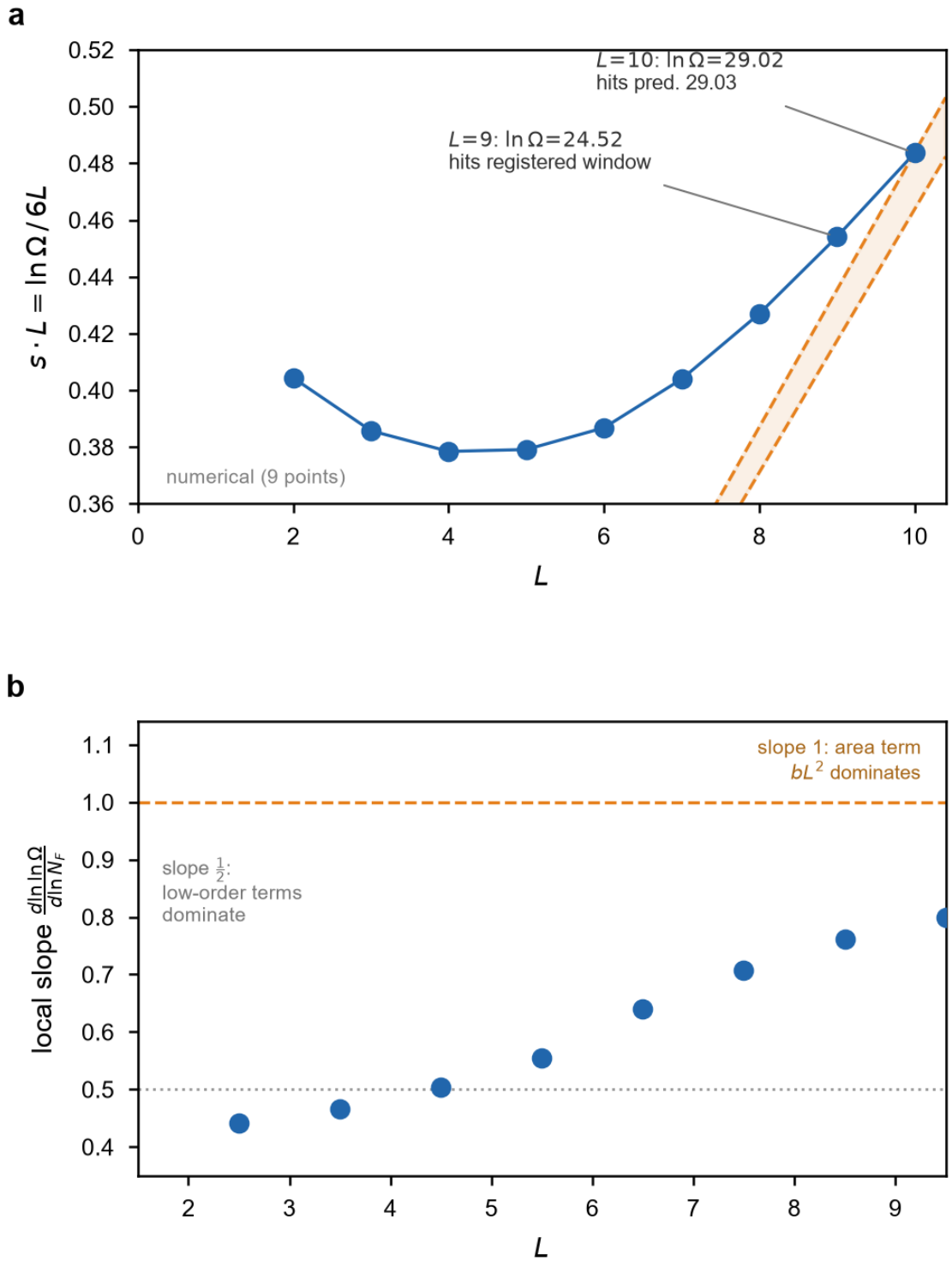



**Fig. S6 | 3.4.6.4 area law. a**, s·L vs L with the registered asymptotic band; the L = 9 and L = 10 points land inside it. **b**, Local log–log slope d ln(ln Ω)/d ln $n_F$ rising monotonically from 0.44 (L = 2–3) toward the extensive value 1: the crossover from sub-extensive transients to area-law scaling.

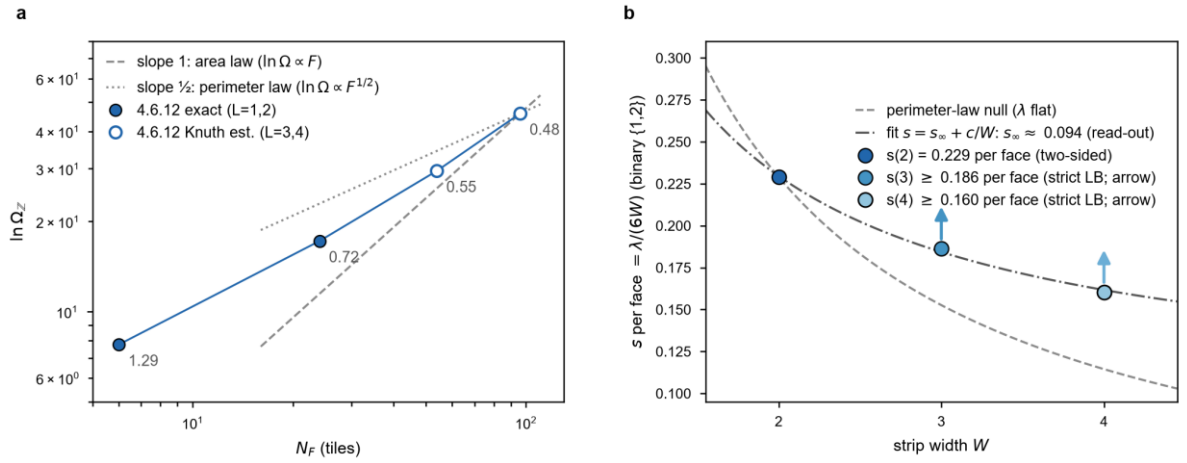



**Fig. S7 | 4.6.12: why the counting series misleads, and what replaces it. a**, ln $\Omega_{\mathbb{Z}}$ versus $N_F$ ({1,2,3} alphabet, L×L tori) with area/perimeter guides. Filled: exact counts (L = 1, 2); open: Knuth tree-sampling estimates (L = 3, 4, anchored at L = 1, 2). Through L = 4 the series runs above the area law and below the perimeter law; the width extrapolation of this series is superseded by the spectral determination in **b** (S8.2). **b**, The spectral determination in the width-scaling language of Fig. 3e: per-face entropy s(W) = λ(W)/(6W) against strip width W. s(2) = 0.229 per face (two-sided); s(3) ≥ 0.186 and s(4) ≥ 0.160 per face (strict lower bounds from trunk-restricted Perron on true-head closures; upward arrows), each strictly above the perimeter-law null s = 2·s(2)/W (dashed) drawn through the W = 2 point. Dash-dotted: a least-squares read-out fit of s = s∞ + c/W (s∞ = 0.094), a model-dependent indication, **not a bound** (S8.2). The width ladder stops at W = 4: W = 5 hits method walls on both routes (S8.2; Reproducibility Supplement).

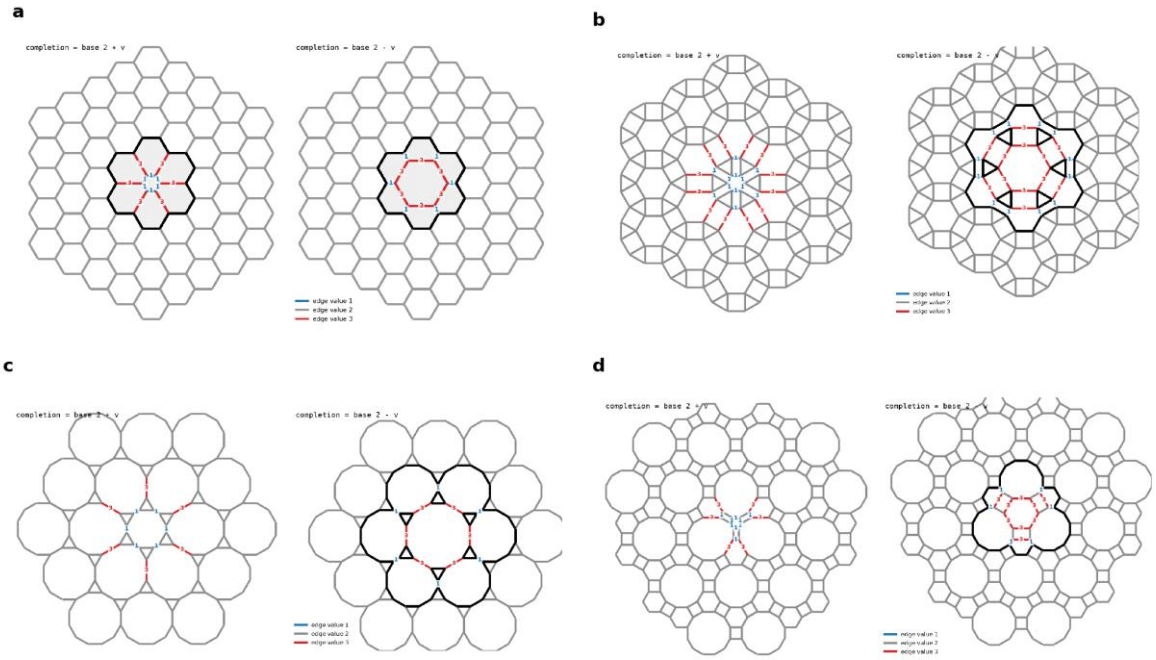



**Fig. S8 | Constructive lower-bound certificates (Theorems S8.3.2–S8.3.3).** Real-metric renderings (edge length = edge value; numeric labels mark the values departing from the base) of the two admissible completions of one certificate patch per carrier, alphabet {1,2,3}, base 2: **a**, honeycomb, the seven-face flower pair (left: $\varphi_+$, the central hexagon contracted to edge 1 with radial edges 3; right: $\varphi_- = 4 - \varphi_+$); **b**, 3.4.6.4 (19 faces: seven hexagons, twelve squares); **c**, 3.12.12, twelve-support-class mid-edge patch; **d**, 4.6.12 (ten faces: three dodecagons, three squares, four hexagons). Disjoint copies of either completion multiply independently (Lemma S8.3.1), giving the per-face bounds (ln 2)/9, (ln 2)/24, (ln 2)/18 and (ln 2)/36 at every system size (S8.3).

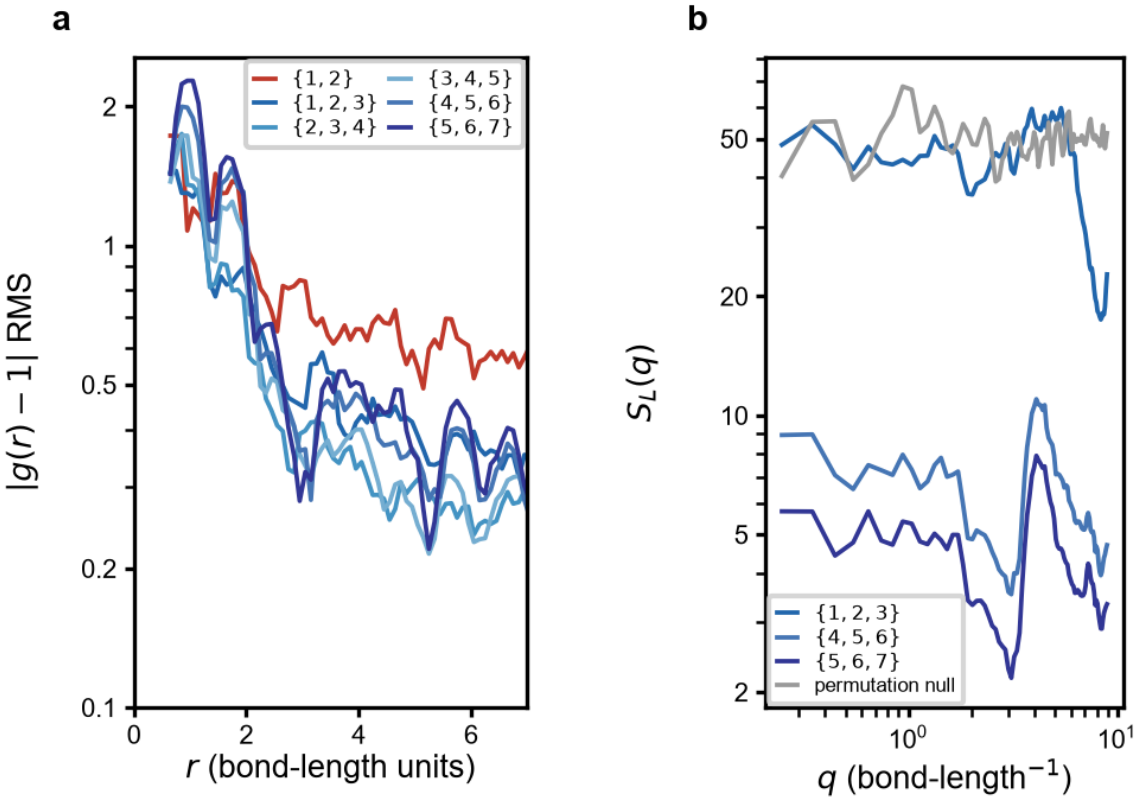



**Fig. S9 | Real-space and reciprocal-space signatures of the delivered suite. a**, The oscillation RMS of g(r)−1 (sliding window; ensemble average over the 20 configurations of each value domain): the envelope falls monotonically and reaches a far-window plateau of 0.28–0.57, with a power-law envelope favoured over an exponential in every domain (S10.1). **b**, $S_L$(q) of the edge-length field (radial average, bright spot cores masked; one permutation null in grey): the null is flat, the arithmetic domains show a mid-q enhancement and a suppressed diffuse floor at the Bragg shells, and the binary domain carries a low-q enhancement (S10.2).

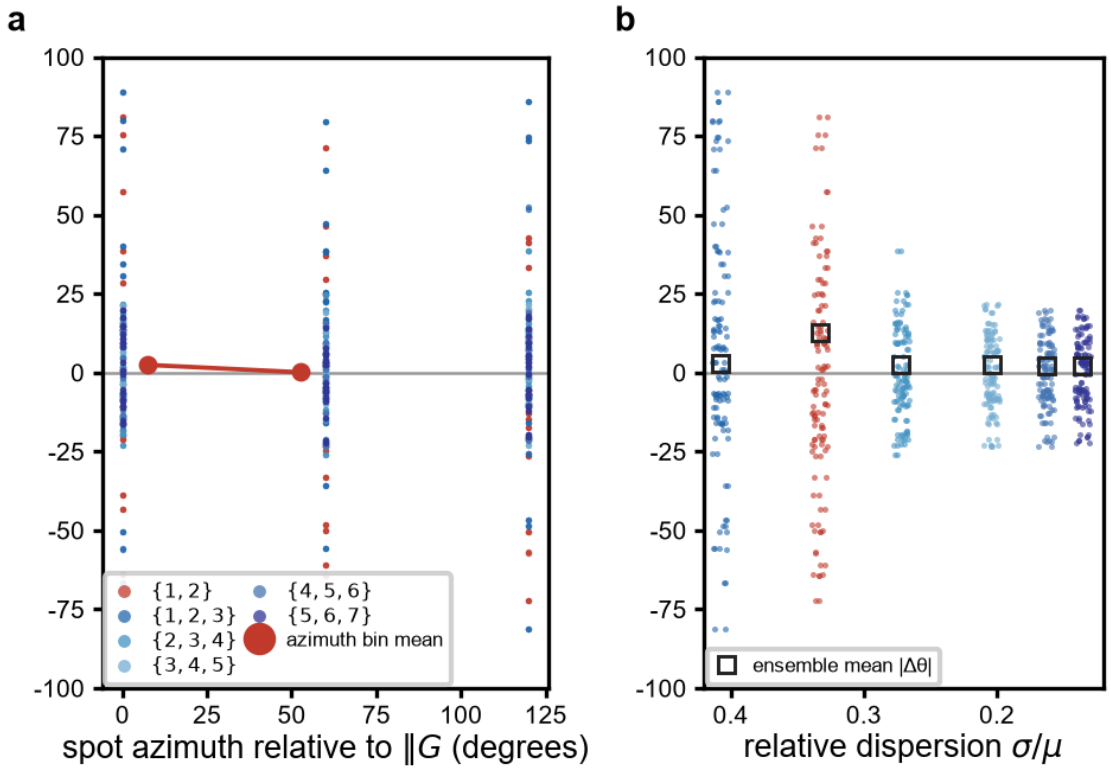



**Fig. S10 | Halo orientation: single-configuration scatter and the ensemble lock. a**, Principal-axis offset Δθ (relative to ‖G) of each spot's diffuse cloud vs spot azimuth, over all 720 configuration×spot samples of the delivered suite (six domains × 20 configurations × 6 spots), coloured by value domain; the scatter fills the range at every azimuth (pooled bin means, red, consistent with zero; no azimuthal structure), while the per-domain spread shrinks along the ladder. **b**, The same Δθ against the domain's relative dispersion σ/μ: per-configuration samples (dots; per-domain s.d. 10.7°–38.6°) and the ensemble-averaged lock |Δθ| (open squares; 2.2–2.9° across the arithmetic domains, 12.7° for the binary). Bragg positions and diffuse orientation belong to the ensemble; the single-configuration scatter belongs to the disorder (S10.3; main text Fig. 4c).

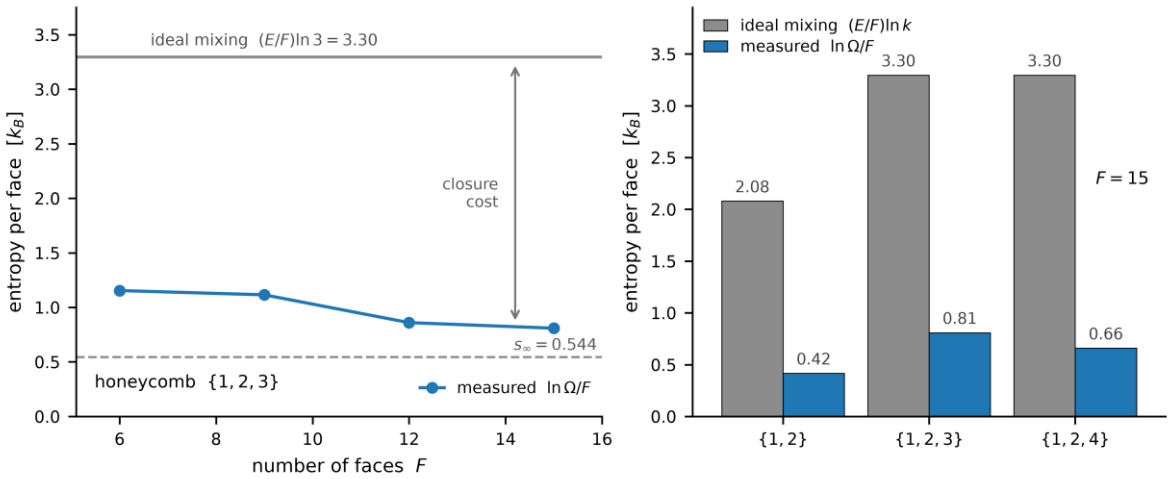



**Fig. S11 | Mixing versus metric: the entropy split at fixed composition. a**, The ternary honeycomb: measured entropy per face lnΩ/F against the counting domain F (blue; exact enumeration, points at F = 6, 9, 12, 15), the ideal alloy-mixing line (E/F)ln 3 = 3.30 (grey), and the transfer-matrix asymptote s∞ = 0.544 (dashed; main text Fig. 3e).

The measured series approaches the asymptote from above; the gap to the ideal-mixing line is the closure cost of the metric constraint, and it grows with the domain (2.14 at F = 6 to 2.49 at F = 15). **b**, The matched comparison at F = 15: the three base alphabets, each against its own ideal-mixing line 3 ln k (2.08 for k = 2, 3.30 for k = 3; grey). {1,2,3} and {1,2,4} share k = 3 and the same ideal-mixing value, yet measure 0.81 against 0.66; the difference is arithmetic, not compositional. The split $\ln\Omega$ = H_comp + $\langle \ln \Omega_N \rangle$ (composition-distribution entropy + conditional count at fixed composition) is exact at every enumerated point; see S11. Data: `si_entropy_split.csv`.

# Tables

**Table S1 | Disambiguation register (theory-lineage neighbours).** Each neighbouring object is separated by a structural criterion, measured where the data allow; the paracrystal comparison is not simulated here (S10.2), so that row is structural rather than measured. Reference numbers refer to the supplementary reference list at the end of this SI.

| State | Disordered DOF | Diffraction | PDF | $\psi_6$ | Entropy | Discriminating criterion |
|---|---|---|---|---|---|---|
| Metric random tiling (this work) | metric (edge lengths = Wang colours) | spots + phason halos | finite decay | ≡ 1 (analytic) | $s\infty > 0$, quantified | — |
| Crystal | none | sharp spot array | no decay | ≡ 1 | 0 | peak height following the size-determined bound ($\alpha \approx 2$) rather than the halo-suppressed scaling of the metric ensemble (S10) |
| Amorphous network | connectivity | continuous ring / speckle | collapsed in 1–2 $l_0$ | < 1 | — | no finite length alphabet (lengths continuous) |
| Paracrystal-type jitter[S14] | metric + positional (coupled); continuous lattice-vector distribution, average local orientation | broadened spots, width does not sharpen with size | decays (different mechanism) | < 1 (angular fluctuations; average frame only) | not countable (continuous distribution) | $S_L(q)$ metric field + discrete motif library (structural; paracrystal ensemble not simulated, Sect. S10.2) |
| Tutte squarings / electrical networks[S15] | metric (side lengths) under fixed right angles | — | — | — | counted for specific dissection problems | single dissection instances (existence and minimal-order problems) versus the entropy density of a periodic closure system over a value domain |

| State | Disordered DOF | Diffraction | PDF | $\psi_6$ | Entropy | Discriminating criterion |
|---|---|---|---|---|---|---|
| Angle-constrained frameworks (angle rigidity)[S16] | directional (angles fixed, lengths free) | — | — | — | dimension/rank criteria only | rank criteria count dimensions, not integer assignments ($\mathbb{Z}$-thinness layer; main-text Sect. 1) |
| Random Regge triangulations[S17] | edge lengths (continuous) on a fluctuating lattice | — | — | — | finite-length ensemble | continuous lengths, variable connectivity, no per-face closure constraint |
| Partial-matching-rule superlattices[S18] | tile arrangement (particle identity = colour) | quasicrystalline spots | — | — | not the counted variable | colours fixed by particle species, not by length; no metric DOF |

$\psi_6$ in this table is the embedded bond-orientational field: $\equiv 1$ means every edge lies on a fixed direction set (crystal, metric random tiling); $<1$ means the local orientation fluctuates (paracrystal with angular fluctuations, amorphous network). The network-topological value is 1 whenever the connectivity is fixed; it is not the discriminator.

**Table S2 | Metric entropy of the eleven computed equiangular networks.** $s_\infty$ in $k_B$ per face; "ext." = extensive; [T] = theorem, [N] = numerical determination with stated calibre, [strict] = mathematically strict bound. The main text cites this catalogue directly (Table S2).

| Network (vertex type) | z | $dim_{\mathbb{R}}$ | $\mathbb{Z}$ / finite-library count | Mechanism | $s_\infty$ | Status |
|---|---|---|---|---|---|---|
| Square $4^4$ | 4 | Lx+Ly (torus) | $\Omega = k^{Lx+Ly}$ exact | propagation | 0 | [T] (S3) |
| Triangular $3^6$ | 6 | 0 | $\Omega = k$ exact | overdetermined rigidity | 0 | [T] (S3) |
| Kagome 3.6.3.6 | 4 | Lq+Lr+gcd−1 | $\Omega \le k^{dim}$ | spectral | 0 | [T] (S4) |
| Honeycomb $6^3$ | 3 | F+2 (ext.) | winding-sector dichotomy: coprime counts of the binary and non-arithmetic classes follow the exact perimeter law $\Omega = k^{Lx+Ly+1}$ (bitwise); resonant branches ride the transfer-matrix Perron asymptote $\lambda(Ly)\cdot Lx$ (S6.3); $\lambda(W)$ grows at every width $\Longrightarrow$ all three classes extensive: $s_\infty = 0.544$ per tile (arithmetic); $\approx 0.2$ for binary and non-arithmetic (common fit band 0.17–0.27; mixed global model, S6.2) | deficiency (extensive kernel) | 0.544 pinned (arithmetic); both other classes $\approx 0.2$ within a fit band 0.17–0.27 [N]; constructive [T]: arithmetic $\ge$ 0.077, binary $\ge$ 0.055 (S8.3) | TM spectra [N] + constructive bounds [T] (S6, S8.3) |

| Network (vertex type) | z | $dim_{\mathbb{R}}$ | $\mathbb{Z}$ / finite-library count | Mechanism | $s_\infty$ | Status |
|---|---|---|---|---|---|---|
| 4.8.8 | 3 | F+2 (ext.) | $\mathbb{Z}$: $\Omega = 3^{4L}$ exact; $\mathbb{Q}(\sqrt{2})$ lattice library: counts sub-extensive (S5) | split rows | 0 [$\mathbb{Z}$]; ≈ 0 [$\Lambda$, sub-extensive] | [T] + counts [N] (S5) |
| 3.12.12 | 3 | F+2 (ext.); $dim_{\mathbb{Q}} = L^2+3L+1$ (quadratic; exact, L = 1…6) | split rows partition the constraints: cut sector triangle-locked (sub-extensive), mid-edge sector extensive (= honeycomb deficiency); transfer-matrix $\lambda(W)$ grows at every width | split rows ($\sqrt{3}/2$); entropy rides the mid-edge carrier | $s_2 \approx 0.13$, $s_3 \approx 0.19$, $s_4 \approx 0.26$–$0.37$ per face [N]; ternary constructive ≥ 0.039 [T] (S8.3) | TM spectra [N] + [T] (S5.3, S8.3) |
| 3.4.6.4 | 4 | $L^2+3$ (ext.) [T] | area law, nine exact points L = 2…10 (S7) | deficiency (triangle-locked) | 0.040 (protocol span 0.031–0.041); ternary constructive ≥ 0.029 [T] (S8.3) | [N] + [T] (S7, S8.3) |
| 4.6.12 | 3 | F+2 (ext.) | $\mathbb{Q}(\sqrt{3})$ split: split-system dim $4L^2+4$ (ext.); binary perimeter law falsified by transfer-matrix spectral rates: rate(3) ≥ 3.351903 and rate(4) ≥ 3.849459, each > 2.747205 = rate(2), trunk-verified (S8.2) | split rows: extensive split lattice; zero-entropy perimeter law falsified | $s_2 \geq 0.186$ per face at W = 3, ≥ 0.160 at W = 4 [strict] (S8.2; the asymptote in S8.4 Q3); ternary constructive ≥ 0.019 [T] (S8.3) | [N]/[strict] + [T] (S8.2, S8.3) |
| 3.3.4.3.4 (snub square) | 5 | 2 exact | $\Omega = k^2$ | triangle-pair rigidity | 0 | [T] (S8) |
| 3.3.3.4.4 | 5 | 2L+1 (sub-ext.) | — | deficiency | 0 | [N] |
| 3.3.3.3.6 (snub hexagonal) | 5 | 1 exact | $\Omega \leq k$ | triangle-locking propagation | 0 | [T]+[N], k = 1…5 (S8) |

**Table S3 | The four mechanisms of metric entropy.** $s_\infty$ in units of $k_B$ per face; [T] = theorem, [N] = numerical determination (calibres in SI). (The main-text table; moved to the SI 2026-10-17.)

| Mechanism | Representative | Criterion | $s_\infty$ |
|---|---|---|---|
| Propagation | square: $\Omega = k^{Lx+Ly}$; Penrose: $\Omega = k^{strips}$ | perimeter law [T] | 0 [T] |
| Spectral | kagome: det $\hat{M}$ zero lines, dim ker = Lq+Lr+gcd−1 | $\Omega \leq k^{dim}$ [T] | 0 [T] |
| Split rows | 4.8.8: $\Omega = 3^{4L}$ on $\mathbb{Z}$; $dim_A = 2\ dim_{\mathbb{R}}$ on $\mathbb{Q}(\sqrt{2})$; 3.12.12: cut sector triangle-locked, mid-edge sector = honeycomb deficiency (SI S5.3); 4.6.12: split lattice extensive (SI S8.2) | arithmetic splitting [T]; lethal when the split rows exhaust the constraints (4.8.8); when they do not, the entropy rides the mid-edge sector (3.12.12) or the extensive split lattice (4.6.12) | 4.8.8: 0 on $\mathbb{Z}$ [T], sub-extensive on $\mathbb{Q}(\sqrt{2})$ [N]; 3.12.12: > 0 [N] (positive at every accessed width, W = 2–8 across k = 2–4, SI S5.3), ternary constructive ≥ (ln 2)/18 [T] (SI S8.3); 4.6.12: $s_2 \geq 0.186$ at width 3 [strict] (perimeter law falsified at two independent widths, W = 3 and W = 4; SI S8.2), ternary constructive ≥ (ln 2)/36 [T] (SI S8.3) |

| Mechanism | Representative | Criterion | $s_\infty$ |
|---|---|---|---|
| Rigidity and deficiency | deficient: honeycomb (0.544 arithmetic; ≈0.2 for the two non-arithmetic classes, pure-metric), 3.4.6.4; rigid or sub-extensively deficient: 3.3.4.3.4, 3.3.3.4.4, 3.3.3.3.6 | kernel of the closure system: extensive families of constraint-row dependencies ("one hexagon, one module", $\mathbb{R}$ layer) → entropy survives; zero or sub-extensive kernel (rigid) → 0; magnitude set by $\mathbb{Z}$-counting | honeycomb > 0 in every alphabet class [N], constructive [T] ≥ 0.077 (ternary) and ≥ 0.055 (binary) (SI S8.3); 3.4.6.4 = 0.040 [N] (binary; equal-weight protocol 0.031, SI S7), constructive ≥ 0.029 [T] (SI S8.3); rigid side: 0 [T]/[N] (Table S2) |

**Table S4 | Transfer-matrix spectral radii and the mixed global model.** $\lambda(L_y) = \ln \rho(L_y)$, per column (Perron roots; even-width values are spectral moduli: ±-pair or root-of-unity families, S6.3). Asymptotes from the mixed global fit $\lambda/L_y = s + c/L_y + A'q^{L_y}$ (binary: even/odd branch lines sharing s, c, q); bands and superseded readings in S6.2.

| Class | $L_y = 2$ | 3 | 4 | 5 | 6 | 7 | 8 | 9 | $s_\infty$ (mixed) |
|---|---|---|---|---|---|---|---|---|---|
| binary {1,2} (k=2) | 0.89588 | 0.96657 | 1.24245 | 1.32452 | 1.59197 | 1.73615 | 1.99572 | 2.17424 | **0.209** (A′ ratio pinned; 0.248 free) |
| non-arithmetic {1,2,4} (k=3) | 1.24245 | 1.38629 | 1.50444 | 1.58305 | 1.78945 | 1.89771 | — | — | **0.224** (free fit; same band as 0.209) |
| arithmetic {1,2,3} (k=3) | 1.60944 | 2.01215 | 2.50211 | 3.01488 | 3.54934 | 4.09316 | 4.643 | — | **0.544** |

**Table S5 | 3.4.6.4 exact torus counts (binary integer alphabet {1,2}, translation-equivalent to {4,5}; see S7.3; L×L tori, F = 6L²).**

| L | Ω | ln Ω | s·L = ln Ω/(6L) |
|---|---|---|---|
| 2 | 128 | 4.8520 | 0.4043 |
| 3 | 1 036 | 6.9431 | 0.3857 |
| 4 | 8 800 | 9.0825 | 0.3784 |
| 5 | 86 956 | 11.3732 | 0.3791 |
| 6 | 1 114 700 | 13.9241 | 0.3868 |
| 7 | 23 325 388 | 16.9651 | 0.4039 |
| 8 | 796 865 888 | 20.4962 | 0.4270 |
| 9 | 44 571 806 560 | 24.5204 | 0.4541 |
| 10 | 4 002 790 772 868 | 29.0178 | 0.4836 |

**Table S6 | 4.6.12 $\mathbb{Z}$ counts and split-lattice ranks** ({1,2,3} alphabet; L×L tori, $N_F = 6L^2$). Ranks exact (Fraction RREF + float-SVD cross-check; pipeline validated against the 3.12.12 dimensions 11/19). $\Omega_{\mathbb{Z}}$ at L = 1, 2 exact (brute force / parametrised DFS); at L = 3, 4 Knuth tree-sampling estimates (final estimator version, anchored at L = 1, 2 against the exact totals, 2 346/2 349 and $2.971 \times 10^7$/2.972 351 7 × $10^7$; 160 k / 2 M samples).

| L | $N_F$ | $dim_{\mathbb{R}}$ | $dim_{split}$ | $\Omega_{\mathbb{Z}}$ | $s_{\mathbb{Z}} = \ln \Omega/N_F$ | method |
|---|---|---|---|---|---|---|
| 1 | 6 | — (degenerate)* | — | 2 349 | 1.294 | exact (brute force) |
| 2 | 24 | 26 = F+2 | 20 = 4L²+4 | 29 723 517 | 0.717 | exact (parametrised DFS) |
| 3 | 54 | 56 = F+2 | 40 = 4L²+4 | ≈ 6.57 × $10^{12}$ | 0.5465 ± 0.0005 | Knuth 160 k |
| 4 | 96 | 98 = F+2 | 68 = 4L²+4 | ≈ 8.35 × $10^{19}$ | 0.4778 ± 0.0001 | Knuth 2 M |

(* = the L = 1 torus is degenerate (equations merge at L = 1) and is used only as the brute-force anchor; the $4L^2+4$ split dimension is exact at L = 2…4, S8.2.)

**Table S7 | The value-domain ensemble suite.** N = 169 holes (13×13 torus, cut open); 20 configurations per domain; exact-uniform two-stage sampler (PT, seed family 20261100 + k, per-configuration seeds archived); f_coh = $|\langle F\rangle|^2/H_{ref}$ ($H_{ref}$: the ideal-patch reference in the same raw-sum estimator, S9 estimators), the halo-to-core ratio and r are six-spot means with configuration-bootstrap 95% intervals (B = 200; per-spot tables archived, halo/core intervals in the `hc_lo`/`hc_hi` columns; within a domain the six spots split into inequivalent classes, per-spot f_coh spreading by up to a factor of five at the disordered end). f_inc = $\langle|F|^2\rangle/H_{ref}$ − f_coh is the incoherent share at the spot positions. S⊥/S∥ is the halo-window displacement-spectrum ratio, mean ± s.d. over the ten resolved k-bins k = 0.34–1.06 (the innermost bin of the halo window lies below the 13×13 supercell's fundamental Bloch shell and is excluded; S10.3). The non-arithmetic {1,2,4} row is deferred (S9 ensembles); the {1,2} row carries the ensemble-level systematic of S9 calibre (vi). The last two arithmetic rows ({6,7,8}, {7,8,9}) extend the ladder to the relative dispersion of alloyed frameworks; they are exact translation relabellings of the {1,2,3} sample (no new sampling). The halo-to-core column is a raw-sum ratio of the two windows whose flat-field value is 24.8 (S9 estimators); the coherent fraction is size-dependent at every domain, most strongly at the disordered end (S10.5). Tension block: the biased ensembles of the arithmetic domain {1,2,3} (metric equation of state m(h) and entropy cost Δs(h), S9 h-ensembles; Fig. 4e–g); every tension is sampled twice (directly and through the mirror of the opposite-tension ensemble) and the quoted f_coh, halo/core and r are the mean of the two samplings with their difference as the spread (S9 calibre vii).

| Domain | Class | Tiles | σ/μ | f_coh | f_inc | halo/core | r | S⊥/S∥ |
|---|---|---|---|---|---|---|---|---|
| {1,2} | binary | 10 | 0.333 | $1.8\times10^{-4}$ | $1.8\times10^{-3}$ | 18.9 | 0.36 | 0.17 ± 0.06 |
| {1,2,3} | arithmetic | 45 | 0.408 | $9.0\times10^{-4}$ | $5.6\times10^{-3}$ | 16.7 | 0.29 | 0.14 ± 0.05 |
| {2,3,4} | arithmetic | 45 | 0.272 | $2.5\times10^{-2}$ | $3.7\times10^{-2}$ | 4.4 | 0.30 | 0.14 ± 0.05 |
| {3,4,5} | arithmetic | 45 | 0.204 | 0.118 | 0.081 | 2.0 | 0.34 | 0.14 ± 0.05 |
| {4,5,6} | arithmetic | 45 | 0.163 | 0.251 | 0.101 | 1.2 | 0.37 | 0.14 ± 0.05 |
| {5,6,7} | arithmetic | 45 | 0.136 | 0.380 | 0.101 | 0.9 | 0.40 | 0.14 ± 0.05 |
| {6,7,8} | arithmetic | 45 | 0.117 | 0.489 | 0.093 | 0.71 | 0.43 | 0.13 ± 0.04 |
| {7,8,9} | arithmetic | 45 | 0.102 | 0.577 | 0.083 | 0.60 | 0.48 | 0.13 ± 0.04 |

*Tension block (arithmetic domain {1,2,3}):*

| h | m(h) [per face] | Δs(h) [$k_B$/face] | f_coh (pooled ± spread) | halo/core | r |
|---|---|---|---|---|---|
| −0.30 | 8.912 | -0.488 | 0.545 ± 0.00939 | 0.62 | 0.59 |
| −0.15 | 7.112 | -0.083 | 0.00432 ± 0.00285 | 9.02 | 0.30 |
| 0 | 6.008 | 0 | $9.0\times10^{-4}$ | 16.7 | 0.29 |
| +0.15 | 4.915 | -0.082 | 0.000119 ± 6.56e-05 | 21.33 | 0.95 |
| +0.30 | 3.100 | -0.490 | 0.192 ± 0.00701 | 1.42 | 0.32 |

*Each tension is read twice, directly and through the mirror ($l \to 4 - l$) of the opposite-tension ensemble; both are exact-distribution samples of the same measure, so the pooled value is their mean and the spread their difference (S9 calibre vii). The mirror table:*

| measure | direct | mirror-read |
|---|---|---|

| measure | direct | mirror-read |
|---|---|---|
| f_coh(h = +0.30) | 0.185 | 0.199 |
| f_coh(h = +0.15) | 5.34e-05 | 0.000185 |
| f_coh(h = -0.15) | 0.00717 | 0.00147 |
| f_coh(h = -0.30) | 0.555 | 0.536 |